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CERTIFICATE

This is to certify that Ms. Prarthi Hrishit Kothari (Roll No: 115) of M. Sc. Bioinformatics (Part II) has satisfactorily completed the practical for Electives Paper: Molecular Docking, Simulation & QSAR (GNKPSBIEL1P504) for Semester IV course prescribed by the University of Mumbai during the academic year 2024-2025.

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WEBLEM: 1

Introduction to Molecular Simulation

AIM:

Introduction to molecular simulations using GROMACS.

INTRODUCTION:

GROMACS (GRONingen MACHINE for Chemical Simulations) is an engine to perform molecular dynamics simulations and energy minimization. These are two of the many techniques that belong to the realm of computational chemistry and molecular modeling. Computational chemistry is just a name to indicate the use of computational techniques in chemistry, ranging from quantum mechanics of molecules to dynamics of large complex molecular aggregates. Molecular modeling indicates the general process of describing complex chemical systems in terms of a realistic atomic model, with the goal being to understand and predict macroscopic properties based on detailed knowledge on an atomic scale. Often, molecular modeling is used to design new materials, for which the accurate prediction of physical properties of realistic systems is required.

Macroscopic physical properties can be distinguished by

1. static equilibrium properties, such as the binding constant of an inhibitor to an enzyme, the average potential energy of a system, or the radial distribution function of a liquid, and
2. dynamic or non-equilibrium properties, such as the viscosity of a liquid, diffusion processes in membranes, the dynamics of phase changes, reaction kinetics, or the dynamics of defects in crystals.

The choice of technique depends on the question asked and on the feasibility of the method to yield reliable results at the present state of the art. Ideally, the (relativistic) time-dependent Schrödinger equation describes the properties of molecular systems with high accuracy, but anything more complex than the equilibrium state of a few atoms cannot be handled at this ab initio level. Thus, approximations are necessary; the higher the complexity of a system and the longer the time span of the processes of interest is, the more severe the required approximations are. At a certain point (reached very much earlier than one would wish), the ab initio approach must be augmented or replaced by empirical parameterization of the model used. Where simulations based on physical principles of atomic interactions still fail due to the complexity of the system, molecular modeling is based entirely on a similarity analysis of known structural and chemical data. The QSAR methods (Quantitative Structure-Activity Relations) and many homology-based protein structure predictions belong to the latter category.

Macroscopic properties are always ensemble averages over a representative statistical ensemble (either equilibrium or non-equilibrium) of molecular systems. For molecular modeling, this has two important consequences:

The knowledge of a single structure, even if it is the structure of the global energy minimum, is not sufficient. It is necessary to generate a representative ensemble at a given temperature,

in order to compute macroscopic properties. But this is not enough to compute thermodynamic equilibrium properties that are based on free energies, such as phase equilibria, binding constants, solubilities, relative stability of molecular conformations, etc. The computation of free energies and thermodynamic potentials requires special extensions of molecular simulation techniques.

While molecular simulations, in principle, provide atomic details of the structures and motions, such details are often not relevant for the macroscopic properties of interest. This opens the way to simplify the description of interactions and average over irrelevant details. The science of statistical mechanics provides the theoretical framework for such simplifications. There is a hierarchy of methods ranging from considering groups of atoms as one unit, describing motion in a reduced number of collective coordinates, averaging over solvent molecules with potentials of mean force combined with stochastic dynamics, to mesoscopic dynamics describing densities rather than atoms and fluxes as response to thermodynamic gradients rather than velocities or accelerations as response to forces. For the generation of a representative equilibrium ensemble two methods are available:

1. Monte Carlo simulations and
2. Molecular Dynamics simulations.

For the generation of non-equilibrium ensembles and for the analysis of dynamic events, only the second method is appropriate. While Monte Carlo simulations are more simple than MD (they do not require the computation of forces), they do not yield significantly better statistics than MD in a given amount of computer time. Therefore, MD is the more universal technique. If a starting configuration is very far from equilibrium, the forces may be excessively large and the MD simulation may fail. In those cases, a robust energy minimization is required. Another reason to perform an energy minimization is the removal of all kinetic energy from the system: if several “snapshots” from dynamic simulations must be compared, energy minimization reduces the thermal noise in the structures and potential energies so that they can be compared better.

Force Fields in GROMACS:

Force fields are the cornerstone of molecular dynamics (MD) simulations. They provide the mathematical framework that describes the interactions between atoms and molecules in a system. In GROMACS, the choice of force field is critical, as it determines the accuracy and reliability of the simulation results. GROMACS supports a wide range of force fields, each designed for specific types of molecules and systems. Below is a detailed overview of the force fields commonly used in GROMACS, their characteristics, and their applications.

What is a Force Field?

A force field is a collection of mathematical functions and parameters that define the potential energy of a system as a function of its atomic coordinates. The total potential energy (U) of a system is typically expressed as the sum of several terms:

$$U = U_{\text{bond}} + U_{\text{angle}} + U_{\text{dihedral}} + U_{\text{non-bonded}} \quad U = U_{\text{bond}} + U_{\text{angle}} + U_{\text{dihedral}} + U_{\text{non-bonded}}$$

- **Bonded Interactions:**

1. U_{bond} : Energy associated with bond stretching (harmonic potential).

2. U_{angle} : Energy associated with angle bending (harmonic potential).
3. U_{dihedral} : Energy associated with torsional rotations (periodic potential).

- **Non-Bonded Interactions:**

1. $U_{\text{non-bonded}}$: Energy associated with van der Waals (Lennard-Jones potential) and electrostatic interactions (Coulomb's law).

Each force field has its own set of parameters for these interactions, which are derived from experimental data and quantum mechanical calculations.

Commonly Used Force Fields in GROMACS

GROMACS supports a variety of force fields, each tailored for specific types of molecules and systems. Below are some of the most widely used force fields in GROMACS:

1. AMBER (Assisted Model Building with Energy Refinement)

- **Description:** The AMBER force field is widely used for simulating proteins, nucleic acids, and other biomolecules. It is known for its accuracy in modeling biomolecular systems.
- **Key Features:**
 - Includes parameters for proteins, DNA, RNA, and carbohydrates.
 - Uses a combination of harmonic potentials for bonds and angles, and periodic functions for dihedrals.
 - Non-bonded interactions are modeled using Lennard-Jones potentials and Coulomb's law.
- **Variants:**
 - **ff99SB:** A widely used variant for proteins, with improved side-chain torsion parameters.
 - **ff14SB:** An updated version with better accuracy for protein simulations.
 - **OL15:** Optimized for RNA simulations.
- **Applications:** Protein folding, nucleic acid dynamics, ligand binding, and free energy calculations.

2. CHARMM (Chemistry at HARvard Macromolecular Mechanics)

- **Description:** The CHARMM force field is another popular choice for biomolecular simulations. It is known for its detailed parameterization and broad applicability.
- **Key Features:**
 - Includes parameters for proteins, lipids, nucleic acids, and carbohydrates.
 - Uses harmonic potentials for bonds and angles, and periodic functions for dihedrals.

- Non-bonded interactions are modeled using Lennard-Jones potentials and Coulomb's law.
- **Variants:**
 - **CHARMM36:** The latest version, with improved parameters for lipids, proteins, and nucleic acids.
 - **CHARMM27:** An older version, still used for specific applications.
- **Applications:** Membrane simulations, protein-ligand interactions, and biomolecular dynamics.

3. OPLS (Optimized Potentials for Liquid Simulations)

- **Description:** The OPLS force field is designed for simulating organic liquids and biomolecules. It is known for its accuracy in modeling small molecules and solvents.
- **Key Features:**
 - Includes parameters for proteins, nucleic acids, and organic molecules.
 - Uses harmonic potentials for bonds and angles, and periodic functions for dihedrals.
 - Non-bonded interactions are modeled using Lennard-Jones potentials and Coulomb's law.
- **Variants:**
 - **OPLS-AA:** All-atom version, suitable for proteins and organic molecules.
 - **OPLS-UA:** United-atom version, which reduces computational cost by treating hydrogen atoms implicitly.
- **Applications:** Solvent simulations, drug design, and protein-ligand interactions.

4. GROMOS (GRoningen MOlecular Simulation)

- **Description:** The GROMOS force field is developed by the University of Groningen and is widely used for biomolecular simulations. It is known for its simplicity and efficiency.
- **Key Features:**
 - Includes parameters for proteins, nucleic acids, and lipids.
 - Uses harmonic potentials for bonds and angles, and periodic functions for dihedrals.
 - Non-bonded interactions are modeled using Lennard-Jones potentials and Coulomb's law.

- **Variants:**
 - **GROMOS 54A7:** The latest version, with improved parameters for proteins and lipids.
 - **GROMOS 53A6:** An older version, still used for specific applications.
- **Applications:** Protein dynamics, lipid bilayers, and biomolecular simulations.

Basic Workflow of a GROMACS Simulation

A typical GROMACS simulation involves several steps, each of which is crucial for obtaining meaningful results. The workflow can be summarized as follows:

1. **System Preparation:** The first step is to prepare the molecular system to be simulated. This involves creating a coordinate file (usually in PDB format) that contains the atomic positions of the molecules in the system. The system may include proteins, lipids, water molecules, ions, and other solutes. GROMACS provides tools like `pdb2gmx` to convert PDB files into GROMACS format and assign appropriate force field parameters.
2. **Solvation and Ionization:** Once the molecular system is prepared, it is typically solvated in a box of water molecules. GROMACS provides tools like `editconf` and `solvate` to place the system in a periodic box and add water molecules. Ions may also be added to neutralize the system or achieve a specific ionic strength.
3. **Energy Minimization:** Before running a molecular dynamics simulation, it is essential to minimize the energy of the system to remove any steric clashes or unfavorable interactions. GROMACS uses steepest descent or conjugate gradient algorithms for energy minimization.
4. **Equilibration:** After energy minimization, the system is equilibrated to bring it to the desired temperature and pressure. This is typically done in two stages: first, a short simulation with position restraints on the solute to allow the solvent to relax around it, followed by a second simulation without restraints to equilibrate the entire system.
5. **Production Run:** Once the system is equilibrated, the production run can begin. This is the main molecular dynamics simulation where the system is allowed to evolve over time according to the laws of classical mechanics. The trajectory of the system is saved at regular intervals for later analysis.
6. **Analysis:** After the simulation is complete, the trajectory is analyzed to extract meaningful information about the system. GROMACS provides a wide range of analysis tools, including `gmx rms` for calculating RMSD, `gmx gyrate` for computing the radius of gyration, and `gmx energy` for extracting energy terms from the simulation.

Why GROMACS?

GROMACS is one of the most popular MD simulation packages due to its combination of **speed**, **accuracy**, and **versatility**. Here are some of the key features that make GROMACS a preferred choice for researchers:

1. **High Performance:** GROMACS is highly optimized for performance, leveraging advanced algorithms and parallel computing techniques to achieve fast simulation speeds. It can efficiently utilize multi-core CPUs and GPUs, making it suitable for simulating large systems or long timescales.
2. **Wide Range of Force Fields:** GROMACS supports multiple force fields, including AMBER, CHARMM, OPLS, and GROMOS, allowing users to choose the most appropriate model for their system. This flexibility is essential for accurately representing different types of molecules and interactions.
3. **Comprehensive Toolset:** GROMACS provides a complete suite of tools for system preparation, simulation, and analysis. These tools simplify the process of building molecular systems, generating input files, and extracting meaningful data from simulations.
4. **Extensive Analysis Capabilities:** GROMACS includes a wide range of analysis tools for computing properties such as root-mean-square deviation (RMSD), radius of gyration, hydrogen bonding, diffusion coefficients, and free energy differences. These tools enable researchers to interpret simulation results and compare them with experimental data.
5. **Active Community and Documentation:** GROMACS has a large and active user community, which contributes to its continuous development and provides support through forums, mailing lists, and tutorials. The software is well-documented, making it accessible to both beginners and experienced users.
6. **Open-Source and Cross-Platform:** GROMACS is open-source and freely available, making it accessible to researchers worldwide. It runs on a variety of platforms, including Linux, macOS, and Windows.

METHODOLOGY:

1. **Update and install dependencies:** Installs required tools like CMake, compilers, and libraries.

sudo apt-get update

sudo apt-get install -y cmake build-essential libfftw3-dev libopenmpi-dev openmpi-bin
2. **Download GROMACS source code:** Fetches the latest stable version of GROMACS.

wget <https://ftp.gromacs.org/gromacs/gromacs-2025.0.tar.gz>
3. **Extract the tarball:** Unpacks the downloaded GROMACS source code.

tar -xzf gromacs-2025.0.tar.gz
4. **Configure and build GROMACS:** Configures the build, compiles GROMACS, and installs it to /opt/gromacs.
 1. **Navigate to the extracted directory:**

```
cd gromacs-2025.0
```

2. Create a build directory and navigate into it:

```
mkdir build
```

```
cd build
```

3. Run CMake to configure the build:

```
cmake.. -DGMX_BUILD_OWN_FFTW=ON  
MAKE_INSTALL_PREFIX=/opt/gromacs
```

(Replace /opt/gromacs with your desired installation directory.)

4. Compile GROMACS using all available CPU cores:

```
make
```

5. Run tests to ensure everything is working (optional but recommended):

```
make check
```

5. Install GROMACS:

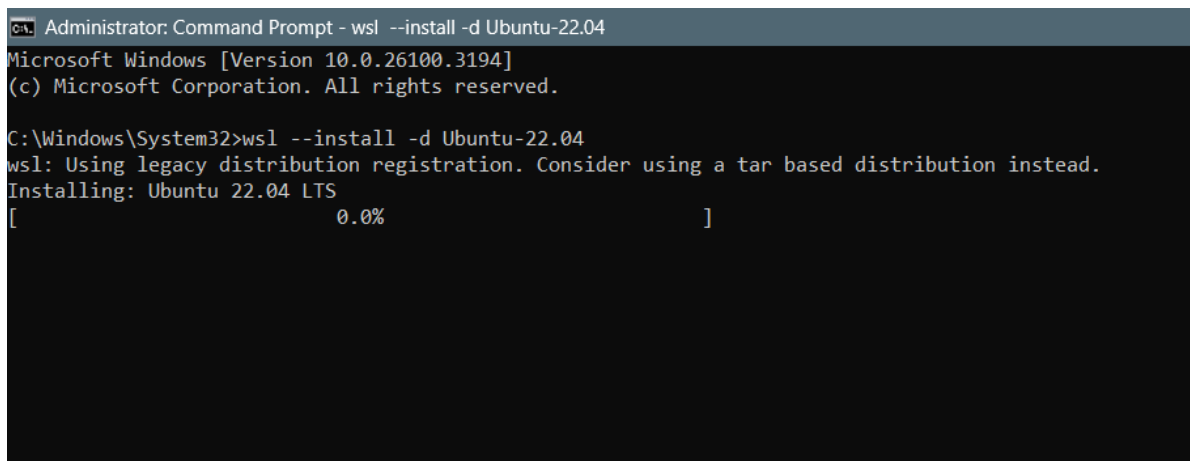
```
sudo make install
```

6. Verify installation: Checks if GROMACS is installed correctly.

```
gmx --version
```

GROMACS INSTALLATION:

STEP 1: INSTALL WSL (WINDOWS SUBSYSTEM FOR LINUX):



```
Administrator: Command Prompt - wsl --install -d Ubuntu-22.04  
Microsoft Windows [Version 10.0.26100.3194]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Windows\System32>wsl --install -d Ubuntu-22.04  
wsl: Using legacy distribution registration. Consider using a tar based distribution instead.  
Installing: Ubuntu 22.04 LTS  
[ 0.0% ]
```

Fig 1: Install Ubuntu 22.04 using command: wsl --install -d Ubuntu-22.04

```
C:\mansea@DESKTOP-GMRN5C6: ~
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>wsl --install -d Ubuntu-22.04
wsl: Using legacy distribution registration. Consider using a tar based distribution instead.
Installing: Ubuntu 22.04 LTS
Ubuntu 22.04 LTS has been installed.
Launching Ubuntu 22.04 LTS...
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: mansea
New password:
Retype new password:
passwd: password updated successfully
Installation successful!
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Mar  9 13:34:47 IST 2025

System load:  0.0               Processes:            51
Usage of /:   0.1% of 1006.85GB  Users logged in:     0
Memory usage: 11%              IPv4 address for eth0: 172.17.200.182
Swap usage:   0%

This message is shown once a day. To disable it please create the
/home/mansea/.hushlogin file.
mansea@DESKTOP-GMRN5C6:~$
```

Fig 2: Ubuntu 22.04 is installed; username and password setup.

Open the downloaded Ubuntu 22.04 and proceed to the process of installing GROMACS 2025.0. The guide referred to is: <https://manual.gromacs.org/documentation/current/install-guide/index.html>

STEP 2: INSTALL C AND C++ COMPILERS:

```
mansea@DESKTOP-GMRN5C6:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2137 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [222 kB]
```

Fig 3: Run command: sudo apt update

```

mansea@DESKTOP-GMRN5C6:~$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bzip2 cpp cpp-11 dpkg-dev fakeroot g++ g++-11 gcc gcc-11 gcc-11-base libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libatomic1 libc-dev-bin libc-devtools libc6 libc6-dev
  libcc1-0 libcrypt-dev libdpkg-perl libfakeroot libfile-fcntllock-perl libgcc-11-dev libgd3 libgomp1 libisl23 libitm
  liblsan0 libmpc3 libnsl-dev libquadmath0 libstdc++-11-dev libtirpc-dev libtsan0 libubsan1 libxpm4 linux-libc-dev
  lto-disabled-list make manpages-dev rpcsvc-proto
Suggested packages:
  bzip2-doc cpp-doc gcc-11-locales debian-keyring g++-multilib g++-11-multilib gcc-11-doc gcc-multilib autoconf
  automake libtool flex bison gdb gcc-doc gcc-11-multilib glibc-doc bsr libgd-tools libstdc++-11-doc make-doc
Recommended packages:
  libnss-nis libnss-nisplus
The following NEW packages will be installed:
  build-essential bzip2 cpp cpp-11 dpkg-dev fakeroot g++ g++-11 gcc gcc-11 gcc-11-base libalgorithm-diff-perl
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0
  libcrypt-dev libdpkg-perl libfakeroot libfile-fcntllock-perl libgcc-11-dev libgd3 libgomp1 libisl23 libitm
  liblsan0 libmpc3 libnsl-dev libquadmath0 libstdc++-11-dev libtirpc-dev libtsan0 libubsan1 libxpm4 linux-libc-dev
  lto-disabled-list make manpages-dev rpcsvc-proto

```

Fig 4: Run command: sudo apt install build-essential

```

mansea@DESKTOP-GMRN5C6:~$ gcc --version && g++ --version
gcc (Ubuntu 11.4.0-1ubuntu1~22.04) 11.4.0
Copyright (C) 2021 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

g++ (Ubuntu 11.4.0-1ubuntu1~22.04) 11.4.0
Copyright (C) 2021 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

```

Fig 5: Check the installation of the compilers: gcc --version && g++ --version

STEP 3: CHECK THAT YOU HAVE CMAKE VERSION 3.28 OR LATER:

```

mansea@DESKTOP-GMRN5C6:~$ cmake --version
Command 'cmake' not found, but can be installed with:
sudo snap install cmake # version 3.31.6, or
sudo apt install cmake # version 3.22.1-1ubuntu1.22.04.2
See 'snap info cmake' for additional versions.

```

Fig 6: Check CMake installation: cmake --version

```

mansea@DESKTOP-GMRN5C6:~$ sudo snap install cmake
[sudo] password for mansea:
snap "cmake" is already installed, see 'snap help refresh'
mansea@DESKTOP-GMRN5C6:~$ cmake --version
cmake version 3.31.6

CMake suite maintained and supported by Kitware (kitware.com/cmake).
mansea@DESKTOP-GMRN5C6:~$ |

```

Fig 7: Install CMake using command: sudo snap install cmake and check installation

STEP 4: DOWNLOADING TARBALL & CREATING BUILD DIRECTORY:

```
mansea@DESKTOP-GMRN5C6:~$ wget https://ftp.gromacs.org/gromacs/gromacs-2025.0.tar.gz
--2025-03-09 15:58:29-- https://ftp.gromacs.org/gromacs/gromacs-2025.0.tar.gz
Resolving ftp.gromacs.org (ftp.gromacs.org)... 130.237.11.165, 2001:6b0:1:1191:216:3eff:fec7:6e30
Connecting to ftp.gromacs.org (ftp.gromacs.org)|130.237.11.165|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 44417653 (42M) [application/x-gzip]
Saving to: 'gromacs-2025.0.tar.gz'

gromacs-2025.0.tar.gz      100%[=====] 42.36M  1.82MB/s
2025-03-09 15:58:57 (1.55 MB/s) - 'gromacs-2025.0.tar.gz' saved [44417653/44417653]
```

Fig 8: Download the latest version of the GROMACS tar file: wget
<https://ftp.gromacs.org/gromacs/gromacs-2025.0.tar.gz>

```
mansea@DESKTOP-GMRN5C6:~$ tar xzf gromacs-2025.0.tar.gz
mansea@DESKTOP-GMRN5C6:~$ cd gromacs-2025.0
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0$ mkdir build
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0$ cd build
```

Fig 9: Unzip the downloaded tar file, change directory to gromacs-2025.0 and make build directory.

1. Unzip the downloaded tar file: `tar xzf gromacs-2025.0.tar.gz`
2. Change directory to GROMACS: `cd gromacs-2025.0`
3. Make new directory: `mkdir build`
4. Move into the new directory: `cd build`

STEP 5: RUN CMAKE WITH THE PATH TO THE SOURCE AS AN ARGUMENT:

```
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0/build$ cmake .. -DGMX_BUILD_OWN_FFTW=ON -DREGRESSIONTEST_DOWNLOAD=ON
-- The C compiler identification is GNU 11.4.0
-- The CXX compiler identification is GNU 11.4.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
```

Fig 10: Run the command: `cmake .. -DGMX_BUILD_OWN_FFTW=ON -DREGRESSIONTEST_DOWNLOAD=ON`

STEP 6: RUN MAKE, MAKE CHECK, AND MAKE INSTALL:

```
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0/build$ make
[ 0%] Creating directories for 'fftwBuild'
[ 0%] Performing download step (download, verify and extract) for 'fftwBuild'
-- Downloading...
   dst='/home/mansea/gromacs-2025.0/build/src/external/build-fftw/fftwBuild-prefix/src/fftw-3.3.10.tar.gz'
   timeout='none'
   inactivity timeout='none'
-- Using src='http://www.fftw.org/fftw-3.3.10.tar.gz'
-- [download 0% complete]
-- [download 1% complete]
-- [download 2% complete]
-- [download 3% complete]
```

Fig 11: Run the command: make

```
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0/build$ make check
[ 0%] Built target gmx_objlib
[ 0%] Built target scanner
[ 1%] Generating release version information
[ 1%] Built target release-version-info
[ 2%] Built target internal_mpi_xdr
```

Fig 12: Run the command: make check

```
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0/build$ sudo make install
[sudo] password for mansea:
[ 1%] Built target fftwBuild
[ 1%] Built target scanner
[ 2%] Generating release version information
[ 2%] Built target release-version-info
[ 2%] Built target internal_mpi_xdr
```

Figure 13: Run the command: sudo make install

STEP 7: SOURCE GMXRC TO GET ACCESS TO GROMACS:

```
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0/build$ source /usr/local/gromacs/bin/GMXRC
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0/build$ gmx
:-) GROMACS - gmx, 2025.0 (-:

Executable: /usr/local/gromacs/bin/gmx
Data prefix: /usr/local/gromacs
```

Fig 13: Run the command: source /usr/local/gromacs/bin/GMXRC

```
mansea@DESKTOP-GMRN5C6:~/gromacs-2025.0/build$ gmx --version
:-) GROMACS - gmx, 2025.0 (-:

Executable: /usr/local/gromacs/bin/gmx
Data prefix: /usr/local/gromacs
Working dir: /home/mansea/gromacs-2025.0/build
Command line:
gmx --version

GROMACS version: 2025.0
Precision: mixed
Memory model: 64 bit
MPI library: thread_mpi
OpenMP support: enabled (GMX_OPENMP_MAX_THREADS = 128)
GPU support: disabled
SIMD instructions: AVX2_256
CPU FFT library: fftw-3.3.10-sse2-avx-avx2-avx2_128
GPU FFT library: none
Multi-GPU FFT: none
```

Fig 15: Check installation using command: gmx --version

Note: If gmx is not recognised run the command *sudo apt install gromacs* and then check the version.

XMGrace:

XMGRACE is a popular open-source software used for 2D plotting and visualization of scientific data. It is particularly useful in molecular simulations, computational chemistry, and bioinformatics, including tasks such as analysing molecular dynamics simulations, visualizing time-series data, and plotting complex datasets.

STEP 1: UPDATE PACKAGE LIST

```
mansea@DESKTOP-GMRN5C6:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for mansea:
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2377 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1193 kB]
Fetched 3954 kB in 4s (1021 kB/s)
Reading package lists... Done
Building dependency tree... Done
```

Fig 16: Run command: *sudo apt update && sudo apt upgrade -y*

STEP 2: INSTALL XMGrace:

```
mansea@DESKTOP-GMRN5C6:~$ sudo apt install grace -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gconf-service gconf-service-backend gconf2 gconf2-common gsfonts libaec0 libice6 libmotif-common libnetcdf19 libsm6 libsz2 libxaw7 libxbae4m libxcb-
  xbitmaps xfonts-100dpi xfonts-encodings xfonts-utils xterm
Suggested packages:
```

Fig 17: Run command: *sudo apt install grace -y*

STEP 3: CHECK IF THE INSTALLATION WAS SUCCESSFUL:

```
mansea@DESKTOP-GMRN5C6:~$ grace -version
Grace-5.1.25

GUI toolkit: @(#)Motif Version 2.3.8
Xbae version: 46004
TlLib: 1.3.1p3-grace
FFT: FFTW
NetCDF support: on
```

Fig 17: Run command: *grace -version*

Note: (For Windows 10 users) XMGrace requires X11 support, which is not enabled by default in WSL as it does not have a built-in GUI; you need an X server to run graphical applications. WSL2 on Windows 11, WSLg (WSL GUI Support) is already included. You don't need XLaunch, as Windows 11 provides a built-in X server.

REFERENCES:

1. Berendsen, H. J. C., van der Spoel, D., & van Drunen, R. (1995). GROMACS: A message-passing parallel molecular dynamics implementation. *Computer Physics Communications*, 91(1-3), 43–56. [https://doi.org/10.1016/0010-4655\(95\)00042-E](https://doi.org/10.1016/0010-4655(95)00042-E)
 2. Abraham, M. J., Murtola, T., Schulz, R., Páll, S., Smith, J. C., Hess, B., & Lindahl, E. (2015). GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX*, 1-2, 19–25. <https://doi.org/10.1016/j.softx.2015.06.001>
 3. Turner, P. J. (2005). Xmgrace. *Computational Science—ICCS 2005*, 3516, 786–791. https://doi.org/10.1007/11428862_101
 4. Pauls, J. (2000). Grace: A tool for visualizing and analyzing scientific data. *Linux Journal*, 2000(78es), 5. <https://www.linuxjournal.com/article/3884>
-

WEBLEM: 1(A)

Molecular Simulation using Gromacs
(URL: <http://www.mdtutorials.com/gmx/>)

AIM:

To demonstrate energy minimization for query “Lysozyme” (PDB ID:1AKI) using GROMACS software.

INTRODUCTION:

Molecular modelling is a collection of computational techniques used to mimic and analyse molecular structures and behaviours at an atomic level. It applies principles from theoretical chemistry, physics, and computer science to predict the properties of molecules and their interactions in various environments. It is widely used in drug design, material science, and structural biology, allowing researchers to visualize molecular conformations, optimize drug candidates, and study biological macromolecules like proteins and nucleic acids.

Energy minimization is a computational process used in molecular modelling to find the most stable (low-energy) conformation of a molecule by reducing steric clashes and unfavourable interactions. It works by adjusting atomic positions iteratively until a local minimum on the potential energy surface is reached. The goal is to obtain a structure that closely represents the molecule's natural state. Methods like Steepest Descent and Conjugate Gradient are commonly used to achieve energy minimization. This process is essential in molecular docking, dynamics simulations, and structural optimization, ensuring that computed models are physically realistic and biologically relevant.

GROMACS (GRoningen MACHine for Chemical Simulations) is a high-performance, open-source software package designed for molecular dynamics simulations of biomolecules such as proteins, lipids, and nucleic acids. It is widely used in computational chemistry and biophysics to simulate the Newtonian equations of motion for large molecular systems, allowing researchers to study biomolecular interactions, conformational changes, and dynamic behaviours. It provides a user-friendly interface, automated topology generation, robust error-checking.

In GROMACS, energy minimization of protein structure is a critical step to refine the protein model by relieving steric clashes, optimizing bond lengths and angles, and stabilizing the molecular geometry before conducting molecular dynamics simulations.

Lysozyme protein- 1AKI:

The 1AKI structure represents the orthorhombic form of hen egg-white lysozyme resolved at 1.5 Å resolution, providing detailed atomic-level insights into its conformation. Lysozyme, a well-studied enzyme, plays a crucial role in hydrolysing the β -(1,4)-glycosidic linkages in bacterial cell walls, making it an essential model protein in structural biology and computational simulations.

METHODOLOGY:

Note: Methodology for energy minimization was referred from the tutorial “Lysozyme in Water”: <http://www.mdtutorials.com/gmx/lysozyme/index.html>

Step 1: Generate Topology

1. Download the protein structure file (PDB Format) -1AKI from RCSB Protein Data Bank: <https://www.rcsb.org/>
2. Visualize the protein structure on Chimera and check the PDB file for missing atoms. If missing atoms/residues are found they must be modeled in using other software packages like Swiss PDB Viewer or Modeller.
3. Open Ubuntu and make a working directory using command: **mkdir Gromacs_Weblem**
4. Copy the downloaded protein 1AKI in the Gromacs_Weblem folder using command: **cp "/mnt/c/Users/Mansi Rawat/Downloads/1aki.pdb" ~/Gromacs_Weblem**
5. Go to the Gromacs_Weblem directory (**cd Gromacs_Weblem**) and remove crystal water from the PDB file using command: **grep -v HOH 1aki.pdb > 1AKI_clean.pdb**
6. Generating the topology file – topol.top using command: **gmx pdb2gmx -f 1AKI_clean.pdb -o 1AKI_processed.gro -water spce**
7. The structure is processed by pdb2gmx, and force field prompt is displayed; the all-atom OPLS force field was chosen, so type 15 at the command prompt, followed by 'Enter'.

Step 2: Solvation

8. Define the box dimensions using the editconf module: **gmx editconf -f 1AKI_processed.gro -o 1AKI_newbox.gro -c -d 1.0 -bt cubic**
9. Fill the box with water using the solvate module command: **gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p topol.top**

Step 3: Add Ions

10. Create an ions file to add ions with .mdp extension (ions.mdp) using: <http://www.mdtutorials.com/gmx/lysozyme/Files/ions.mdp> and copy the file in the Gromacs_Weblem using command: **cp "/mnt/c/Users/Mansi Rawat/Downloads/ions.mdp" ~/Gromacs_Weblem**
11. Assemble the .tpr file using the command: **gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o ions.tpr**
12. Passing the file to genion: **gmx genion -s ions.tpr -o 1AKI_solv_ions.gro -p topol.top -pname NA -nname CL -neutral**
13. Choosing group 13 ‘SOL’ for embedding ions.

Step 4: Energy Minimization

14. Create an input parameter file named “minim” with an extension .mdp (minim.mdp) file using: <http://www.mdtutorials.com/gmx/lysozyme/Files/minim.mdp> and copy the file in the Gromacs_Weblem using command: **cp "/mnt/c/Users/Mansi Rawat/Downloads/minim.mdp" ~/Gromacs_Weblem**
15. Assemble the binary input: **gmx grompp -f minim.mdp -c 1AKI_solv_ions.gro -p topol.top -o em.tpr**
16. Invoke mdrun to carry out the EM using command: **gmx mdrun -v -deffnm em**

17. Analyze any .edr file using the GROMACS energy module: **gmx energy -f em.edr -o potential.xvg** (At the prompt, type "10" to select Potential).
18. Plot the potential energy graph: **xmgrace potential.xvg**
19. Convert the em.gro file to em.pdb for docking using command: **gmx editconf -f em.gro -o em.pdb**
20. Removing the solvation box: **gmx trjconv -s em.tpr -f em.pdb -o em_clean.pdb** It will prompt "Select group for output" → Choose "Protein" (group 1).
21. Copy the em_clean.pdb file to the downloads folder: **cp em_clean.pdb /mnt/c/Users/Mansi\ Rawat/Downloads/** and view on Chimera.

OBSERVATIONS:

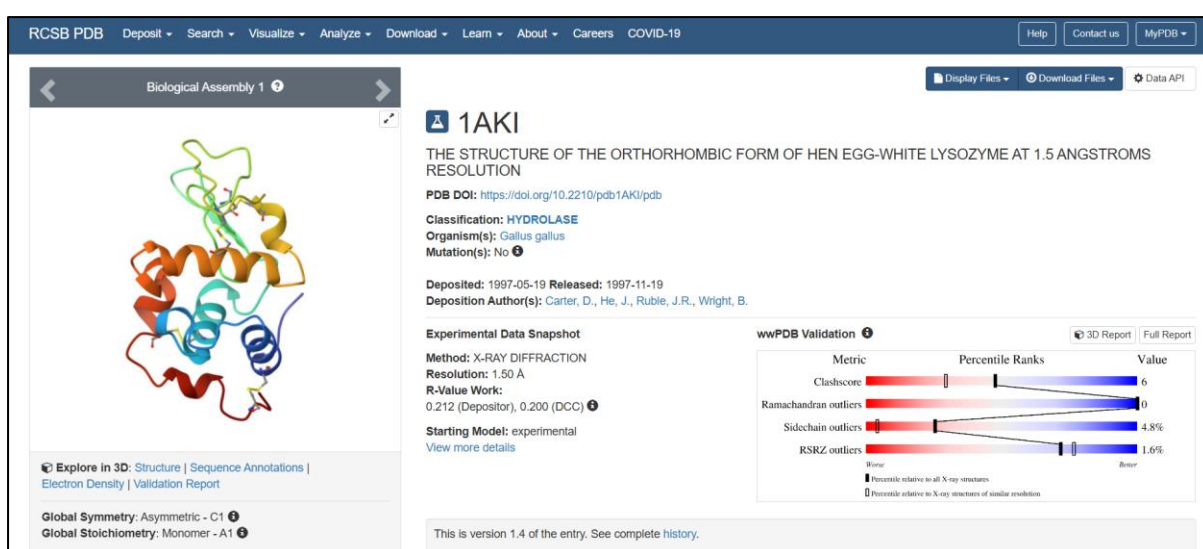


Fig 1: Downloading the protein structure in PDB Format from RCSB PDB

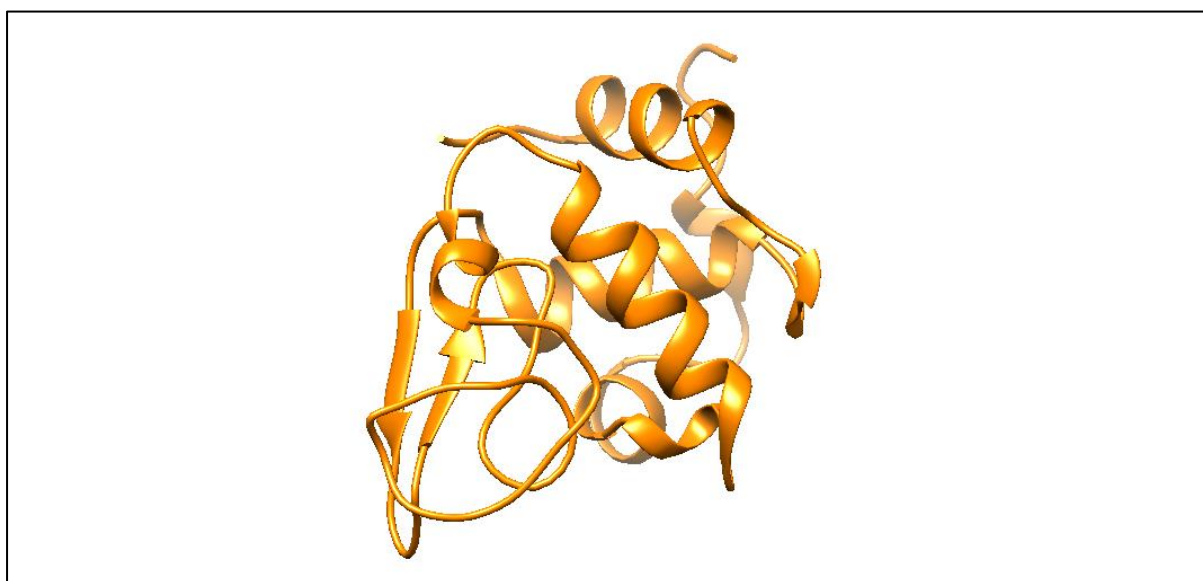


Fig 2: Visualising 1AKI in Chimera

HEADER	HYDROLASE	19-MAY-97	1AKI
TITLE	THE STRUCTURE OF THE ORTHORHOMBIC FORM OF HEN EGG-WHITE LYSOZYME AT		
TITLE	2 1.5 ANGSTROMS RESOLUTION		
COMPND	MOL_ID: 1;		
COMPND	2 MOLECULE: LYSOZYME;		
COMPND	3 CHAIN: A;		
COMPND	4 EC: 3.2.1.17		
SOURCE	MOL_ID: 1;		
SOURCE	2 ORGANISM_SCIENTIFIC: GALLUS GALLUS;		
SOURCE	3 ORGANISM_COMMON: CHICKEN;		
SOURCE	4 ORGANISM_TAXID: 9031;		
SOURCE	5 CELL: EGG		
KEYWDS	HYDROLASE, GLYCOSIDASE		
EXPDTA	X-RAY DIFFRACTION		
AUTHOR	D.CARTER,J.HE,J.R.RUBLE,B.WRIGHT		

Fig 3: Inspecting PDB File for missing residues

```
mansea@DESKTOP-GMRN5C6:~$ cp "/mnt/c/Users/Mansi Rawat/Downloads/1aki.pdb" ~/Gromacs_Weblem
mansea@DESKTOP-GMRN5C6:~$ cd Gromacs_Weblem
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ ls
1aki.pdb
```

Fig 4: Copy the downloaded protein 1AKI in the Gromacs_Weblem folder

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ grep -v HOH 1aki.pdb > 1aki_clean.pdb
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ ls
1aki.pdb 1aki_clean.pdb
```

Fig 5: Running 'grep' command in the Gromacs_Weblem folder to remove crystal water molecules and generate 1aki_clean.pdb file which contains only the protein atoms.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx pdb2gmx -f 1aki_clean.pdb -o 1AKI_processed.gro -water spce
:-) GROMACS - gmx pdb2gmx, 2025.0 (-:
```

Fig 6: Generating the topology file – topol.top using GROMACS module pdb2gmx.

Note: The topology (topol.top by default) contains all the information necessary to define the molecule within a simulation. This information includes nonbonded parameters (atom types and charges) as well as bonded parameters (bonds, angles, and dihedrals).

```
Select the Force Field:

From '/usr/local/gromacs/share/gromacs/top':

 1: AMBER03 protein, nucleic AMBER94 (Duan et al., J. Comp. Chem. 24, 1999-2012, 2003)
 2: AMBER94 force field (Cornell et al., JACS 117, 5179-5197, 1995)
 3: AMBER96 protein, nucleic AMBER94 (Kollman et al., Acc. Chem. Res. 29, 461-469, 1996)
 4: AMBER99 protein, nucleic AMBER94 (Wang et al., J. Comp. Chem. 21, 1049-1074, 2000)
 5: AMBER99SB protein, nucleic AMBER94 (Hornak et al., Proteins 65, 712-725, 2006)
 6: AMBER99SB-ILDN protein, nucleic AMBER94 (Lindorff-Larsen et al., Proteins 78, 1950-58, 2010)
 7: AMBERGS force field (Garcia & Sanbonmatsu, PNAS 99, 2782-2787, 2002)
 8: CHARMM27 all-atom force field (CHARM22 plus CMAP for proteins)
 9: GROMOS96 43a1 force field
10: GROMOS96 43a2 force field (improved alkane dihedrals)
11: GROMOS96 45a3 force field (Schuler JCC 2001 22 1205)
12: GROMOS96 53a5 force field (JCC 2004 vol 25 pag 1656)
13: GROMOS96 53a6 force field (JCC 2004 vol 25 pag 1656)
14: GROMOS96 54a7 force field (Eur. Biophys. J. (2011), 40,, 843-856, DOI: 10.1007/s00249-011-0700-9)
15: OPLS-AA/L all-atom force field (2001 aminoacid dihedrals)
15
```

Fig 7: Selecting a force field; 15 OPLS-AA/L

Note: The force field will contain the information that will be written to the topology.

```

Checking for duplicate atoms...

Generating any missing hydrogen atoms and/or adding termini.

Now there are 129 residues with 1960 atoms

Making bonds...

Number of bonds was 1984, now 1984

Generating angles, dihedrals and pairs...
Before cleaning: 5142 pairs
Before cleaning: 5187 dihedrals

Making cmap torsions...

There are 5187 dihedrals, 426 impropers, 3547 angles
          5106 pairs,      1984 bonds and      0 virtual sites

Total mass 14313.193 a.m.u.

Total charge 8.000 e

Writing topology

Writing coordinate file...

          ----- PLEASE NOTE -----

You have successfully generated a topology from: laki_clean.pdb.

The Oplsaa force field and the spce water model are used.

          ----- ETON ESAELP -----

GROMACS reminds you: "Once Again Let Me Do This" (Urban Dance Squad)

mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ ls
1AKI_processed.gro laki.pdb laki_clean.pdb posre.itp topol.top

```

Fig 8: Information about 1AKI molecule displayed during the run and viewing the files generated.

Note: After all these steps three new files are generated: 1AKI_processed.gro, topol.top, and posre.itp.

1. 1AKI_processed.gro is a GROMACS-formatted structure file that contains all the atoms defined within the force field (i.e., H atoms have been added to the amino acids in the protein).
2. The topol.top file is the system topology.
3. The posre.itp file contains information used to restrain the positions of heavy atom.

```

File 'topol.top' was generated
By user: unknown (1000)
On host: DESKTOP-GMRN5C6
At date: Sun Mar  9 18:50:54 2025

This is a standalone topology file

Created by:
          :-) GROMACS - gmx pdb2gmx, 2025.0 (-:

Executable:  /usr/local/gromacs/bin/gmx
Data prefix:  /usr/local/gromacs
Working dir:  /home/mansea/Gromacs_Weblem
Command line:
          gmx pdb2gmx -f laki_clean.pdb -o 1AKI_processed.gro -water spce
Force field was read from the standard GROMACS share directory.

; Include forcefield parameters
#include "oplsaa.ff/forcefield.itp"

[ moleculetype ]
; Name          nrexcl
Protein_chain_A      3

[ atoms ]
; nr      type  resnr residue  atom   cgnr      charge      mass  typeB      chargeB      massB
; residue  1 LYS rtp LYSH q +2.0
1  opl_287      1  LYS      N       1       -0.3      14.0027
2  opl_290      1  LYS      H1      1        0.33      1.008
3  opl_290      1  LYS      H2      1        0.33      1.008
4  opl_290      1  LYS      H3      1        0.33      1.008
5  opl_293B     1  LYS      CA      1        0.25      12.011
6  opl_140      1  LYS      HA      1        0.06      1.008
7  opl_136      1  LYS      CB      2       -0.12      12.011
8  opl_140      1  LYS      HB1     2        0.06      1.008
9  opl_140      1  LYS      HB2     2        0.06      1.008
10 opl_136      1  LYS      CG      3       -0.12      12.011
11 opl_140      1  LYS      HG1     3        0.06      1.008
12 opl_140      1  LYS      HG2     3        0.06      1.008

```

Fig 9: topol.top file viewed using nano topol.top command.

```

mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx editconf -f 1AKI_processed.gro -o 1AKI_newbox.gro -c -d 1.0 -bt cubic
          :-) GROMACS - gmx editconf, 2025.0 (-:

Executable:  /usr/local/gromacs/bin/gmx
Data prefix:  /usr/local/gromacs
Working dir:  /home/mansea/Gromacs_Weblem
Command line:
          gmx editconf -f 1AKI_processed.gro -o 1AKI_newbox.gro -c -d 1.0 -bt cubic

Note that major changes are planned in future for editconf, to improve usability and utility.
Read 1960 atoms
Volume: 123.376 nm^3, corresponds to roughly 55500 electrons
No velocities found
  system size :  3.817  4.234  3.454 (nm)
  diameter    :  5.010              (nm)
  center      :  2.781  2.488  0.017 (nm)
  box vectors :  5.906  6.845  3.052 (nm)
  box angles  :  90.00  90.00  90.00 (degrees)
  box volume  : 123.38              (nm^3)
  shift       :  0.724  1.017  3.488 (nm)
  new center  :  3.505  3.505  3.505 (nm)
  new box vectors :  7.010  7.010  7.010 (nm)
  new box angles :  90.00  90.00  90.00 (degrees)
  new box volume : 344.48              (nm^3)

GROMACS reminds you: "A Man Needs a Maid" (N. Young)

```

Fig 10: Defining a cubic box around the protein molecule using editconf command.

```

mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ ls
1AKI_newbox.gro  1AKI_processed.gro  laki.pdb  laki_clean.pdb  posre.itp  topol.top

```

Fig 11: Cubic box file: 1AKI_newbox.gro generated.


```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p topol.top
;-) GROMACS - gmx solvate, 2025.0 (-:

Executable: /usr/local/gromacs/bin/gmx
Data prefix: /usr/local/gromacs
Working dir: /home/mansea/Gromacs_Weblem
Command line:
  gmx solvate -cp 1AKI_newbox.gro -cs spc216.gro -o 1AKI_solv.gro -p topol.top

Reading solute configuration
Reading solvent configuration

Initialising inter-atomic distances...

WARNING: Masses and atomic (Van der Waals) radii will be guessed
         based on residue and atom names, since they could not be
         definitively assigned from the information in your input
         files. These guessed numbers might deviate from the mass
         and radius of the atom type. Please check the output
         files if necessary. Note, that this functionality may
         be removed in a future GROMACS version. Please, consider
         using another file format for your input.

NOTE: From version 5.0 gmx solvate uses the Van der Waals radii
      from the source below. This means the results may be different
      compared to previous GROMACS versions.
```

Fig 12: Filling the cubic box with solvent (water) to simulate a simple aqueous system.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ ls
'#topol.top.1#'  1AKI_newbox.gro  1AKI_processed.gro  1AKI_solv.gro  laki.pdb  laki_clean.pdb  posre.itp  topol.top
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$
```

Fig 13: '#topol.top.1#' file generated in the folder.

Note: Solvation step tracks the number of water molecules that are added and writes to the topology to reflect the changes that have been made in the '#topol.top.1#' file. We now have a solvated system that contains a charged protein with a net charge of +8e (based on its amino acid composition).

The tool for adding ions within GROMACS is called genion; which reads the topology and replaces the water molecules with user specific ions. The input is called a run input file with extension .tpr produced by the GROMACS grompp module (GROMACS pre-processor).

The .tpr file contains all the parameters for all of the atoms in the system. To produce a .tpr file with grompp, an additional input file, with the extension .mdp (molecular dynamics parameter file) is needed.

Grompp will assemble the parameters specified in the .mdp file with the coordinates and topology information to generate a .tpr file.

```
; ions.mdp - used as input into grompp to generate ions.tpr
; Parameters describing what to do, when to stop and what to save
integrator      = steep           ; Algorithm (steep = steepest descent minimization)
emtol           = 1000.0          ; Stop minimization when the maximum force < 1000.0 kJ/mol/nm
emstep          = 0.01            ; Minimization step size
nsteps          = 50000           ; Maximum number of (minimization) steps to perform

; Parameters describing how to find the neighbors of each atom and how to calculate the interactions
nstlist         = 1              ; Frequency to update the neighbor list and long range forces
cutoff-scheme   = Verlet         ; Buffered neighbor searching
ns_type         = grid           ; Method to determine neighbor list (simple, grid)
coulombtype     = cutoff         ; Treatment of long range electrostatic interactions
rcoulomb        = 1.0            ; Short-range electrostatic cut-off
rvdw            = 1.0            ; Short-range Van der Waals cut-off
pbc             = xyz            ; Periodic Boundary Conditions in all 3 dimensions
```

Fig 14: Creating an ions.mdp file on notepad with .mdp extension.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ cp "/mnt/c/Users/Mansi Rawat/Downloads/ions.mdp" ~/Gromacs_Weblem
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ ls
'#topol.top.1#' 1AKI_newbox.gro 1AKI_processed.gro 1AKI_solv.gro 1aki.pdb 1aki_clean.pdb ions.mdp posre.itp topol.top
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$
```

Fig 15: Copying the ions.mdp to the Gromacs_Weblem folder.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o ions.tpr
:~) GROMACS - gmx grompp, 2025.0 (-:

Executable: /usr/local/gromacs/bin/gmx
Data prefix: /usr/local/gromacs
Working dir: /home/mansea/Gromacs_Weblem
Command line:
gmx grompp -f ions.mdp -c 1AKI_solv.gro -p topol.top -o ions.tpr

Ignoring obsolete mdp entry 'ns_type'

NOTE 1 [file ions.mdp]:
With Verlet lists the optimal nstlist is >= 10, with GPUs >= 20. Note
that with the Verlet scheme, nstlist has no effect on the accuracy of
your simulation.

Setting the LD random seed to -318777815
```

Fig 16: Assembling a .tpr file containing all atomic-level description of our system in the binary file ions.tpr.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx genion -s ions.tpr -o 1AKI_solv_ions.gro -p topol.top -pname NA -nname CL -neutral
:~) GROMACS - gmx genion, 2025.0 (-:

Executable: /usr/local/gromacs/bin/gmx
Data prefix: /usr/local/gromacs
Working dir: /home/mansea/Gromacs_Weblem
Command line:
gmx genion -s ions.tpr -o 1AKI_solv_ions.gro -p topol.top -pname NA -nname CL -neutral

Reading file ions.tpr, VERSION 2025.0 (single precision)
Reading file ions.tpr, VERSION 2025.0 (single precision)
Will try to add 0 NA ions and 8 CL ions.
Select a continuous group of solvent molecules
Group 0 (System) has 33892 elements
Group 1 (Protein) has 1960 elements
Group 2 (Protein-H) has 1001 elements
Group 3 (C-alpha) has 129 elements
Group 4 (Backbone) has 387 elements
Group 5 (MainChain) has 517 elements
Group 6 (MainChain+Cb) has 634 elements
Group 7 (MainChain+H) has 646 elements
Group 8 (SideChain) has 1314 elements
Group 9 (SideChain-H) has 484 elements
Group 10 (Prot-Masses) has 1960 elements
Group 11 (non-Protein) has 31932 elements
Group 12 (Water) has 31932 elements
Group 13 (SOL) has 31932 elements
Group 14 (non-Water) has 1960 elements
Select a group: 13
Selected 13: 'SOL'
Number of (3-atomic) solvent molecules: 10644

Processing topology
Replacing 8 solute molecules in topology file (topol.top) by 0 NA and 8 CL ions.

Back Off! I just backed up topol.top to ./#topol.top.2#
Using random seed -30146950.
Replacing solvent molecule 5333 (atom 17959) with CL
Replacing solvent molecule 5198 (atom 17554) with CL
Replacing solvent molecule 7399 (atom 24157) with CL
Replacing solvent molecule 10196 (atom 32548) with CL
Replacing solvent molecule 279 (atom 2797) with CL
Replacing solvent molecule 686 (atom 4018) with CL
Replacing solvent molecule 8629 (atom 27847) with CL
```

Fig 17: Passing the ions.tpr file to genion and selecting Group 13 SOL to embed ions.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ ls
'#topol.top.1#' 1AKI_newbox.gro 1AKI_solv.gro 1aki.pdb ions.mdp mdout.mdp topol.top
'#topol.top.2#' 1AKI_processed.gro 1AKI_solv_ions.gro 1aki_clean.pdb ions.tpr posre.itp
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$
```

Fig 18: Listing all the files generated.

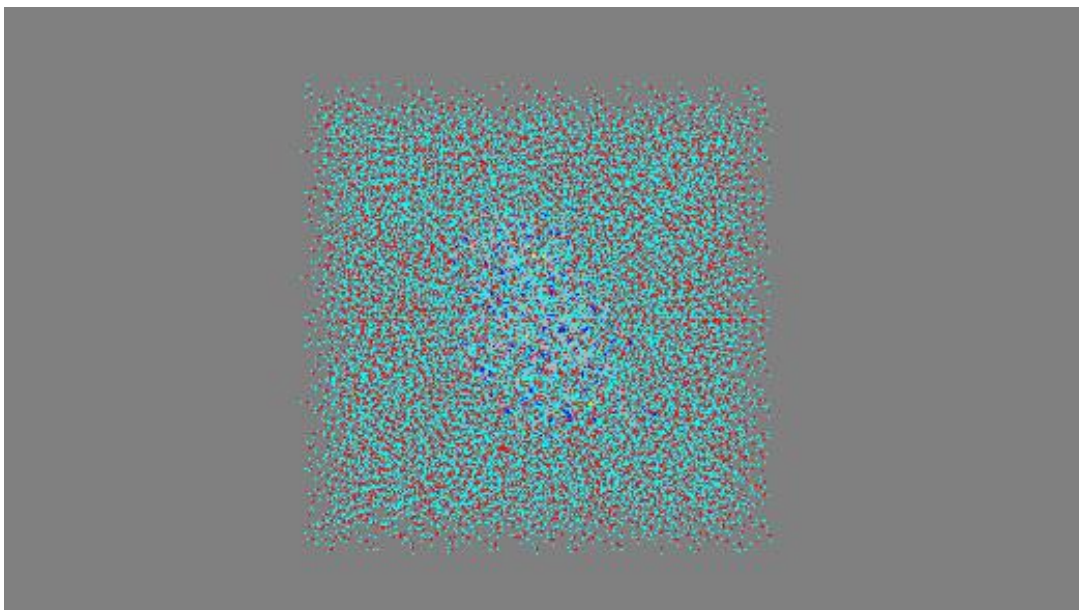


Fig 19: Solvent box around the protein molecule embedded with ions; viewed on PyRx (optional).

Note: These steps assemble a system that is solvated and electroneutral. The structure is relaxed to ensure that the system has no steric clashes or inappropriate geometry through energy minimization (EM). Grompp is used to assemble the structure, topology, and simulation parameters into a binary input file (.tpr) which is passed through the GROMACS MD engine, mdrun for EM. The process requires an input parameter file (.mdp).

```
; minim.mdp - used as input into grompp to generate em.tpr
; Parameters describing what to do, when to stop and what to save
integrator    = steep           ; Algorithm (steep = steepest descent minimization)
emtol         = 1000.0          ; Stop minimization when the maximum force < 1000.0 kJ/mol/nm
emstep        = 0.01           ; Minimization step size
nsteps        = 50000          ; Maximum number of (minimization) steps to perform

; Parameters describing how to find the neighbors of each atom and how to calculate the interactions
nstlist       = 1              ; Frequency to update the neighbor list and long range forces
cutoff-scheme = Verlet         ; Buffered neighbor searching
ns_type       = grid          ; Method to determine neighbor list (simple, grid)
coulombtype   = PME           ; Treatment of long range electrostatic interactions
rcoulomb      = 1.0           ; Short-range electrostatic cut-off
rvdw          = 1.0           ; Short-range Van der Waals cut-off
pbc           = xyz           ; Periodic Boundary Conditions in all 3 dimensions
```

Fig 20: Creating an minim.mdp file on notepad with .mdp extension.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ cp "/mnt/c/Users/Mansi Rawat/Downloads/mini.mdp" ~/Gromacs_Weblem
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ ls
'1topol.top.1#' 1AKI_newbox.gro 1AKI_solv.gro 1aki.pdb ions.mdp mdout.mdp posre.itp
'1topol.top.2#' 1AKI_processed.gro 1AKI_solv_ions.gro 1aki_clean.pdb ions.tpr mini.mdp topol.top
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$
```

Fig 21: Copying the minim.mdp to the Gromacs_Weblem folder.

```

mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx grompp -f mini.mdp -c 1AKI_solv_ions.gro -p topol.top -o em.tpr
:~) GROMACS - gmx grompp, 2025.0 (-:

Executable: /usr/local/gromacs/bin/gmx
Data prefix: /usr/local/gromacs
Working dir: /home/mansea/Gromacs_Weblem
Command line:
  gmx grompp -f mini.mdp -c 1AKI_solv_ions.gro -p topol.top -o em.tpr

Ignoring obsolete mdp entry 'ns_type'

NOTE 1 [file mini.mdp]:
  With Verlet lists the optimal nstlist is >= 10, with GPUs >= 20. Note
  that with the Verlet scheme, nstlist has no effect on the accuracy of
  your simulation.

Setting the LD random seed to -824213572

Generated 330891 of the 330891 non-bonded parameter combinations
Generating 1-4 interactions: fudge = 0.5

Generated 330891 of the 330891 1-4 parameter combinations

Excluding 3 bonded neighbours molecule type 'Protein_chain_A'

Excluding 2 bonded neighbours molecule type 'SOL'

Excluding 1 bonded neighbours molecule type 'CL'
Analysing residue names:
There are: 129      Protein residues
There are: 10636   Water residues
There are: 8       Ion residues
Analysing Protein...
Number of degrees of freedom in T-Coupling group rest is 69717.00
The integrator does not provide a ensemble temperature, there is no system ensemble temperature

```

Fig 22: Assembling the .tpr file for EM – em.tpr.

```

mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx mdrun -v -deffnm em
:~) GROMACS - gmx mdrun, 2025.0 (-:

Executable: /usr/local/gromacs/bin/gmx
Data prefix: /usr/local/gromacs
Working dir: /home/mansea/Gromacs_Weblem
Command line:
  gmx mdrun -v -deffnm em

Reading file em.tpr, VERSION 2025.0 (single precision)
Using 1 MPI thread
Using 4 OpenMP threads

Steepest Descents:
  Tolerance (Fmax) = 1.00000e+03
  Number of steps = 50000
Step= 0, Dmax= 1.0e-02 nm, Epot= -3.31630e+05 Fmax= 2.28196e+05, atom= 1905
Step= 1, Dmax= 1.0e-02 nm, Epot= -3.47697e+05 Fmax= 8.64714e+04, atom= 19883
Step= 2, Dmax= 1.2e-02 nm, Epot= -3.70026e+05 Fmax= 3.30256e+04, atom= 19883
Step= 3, Dmax= 1.4e-02 nm, Epot= -3.98427e+05 Fmax= 1.32029e+04, atom= 24188
Step= 4, Dmax= 1.7e-02 nm, Epot= -4.26949e+05 Fmax= 5.65293e+03, atom= 21860
Step= 5, Dmax= 2.1e-02 nm, Epot= -4.52082e+05 Fmax= 8.78279e+03, atom= 1905
Step= 6, Dmax= 2.5e-02 nm, Epot= -4.58466e+05 Fmax= 2.48909e+04, atom= 1905
Step= 7, Dmax= 3.0e-02 nm, Epot= -4.64122e+05 Fmax= 1.50237e+04, atom= 1905
Step= 8, Dmax= 3.6e-02 nm, Epot= -4.64577e+05 Fmax= 3.45854e+04, atom= 1905
Step= 9, Dmax= 4.3e-02 nm, Epot= -4.69886e+05 Fmax= 2.70383e+04, atom= 112

```

Fig 23: Invoking mdrun to carry out the energy minimisation

```

Step= 830, Dmax= 7.3e-03 nm, Epot= -5.87497e+05 Fmax= 6.86467e+03, atom= 736
Step= 831, Dmax= 8.8e-03 nm, Epot= -5.87423e+05 Fmax= 6.86467e+03, atom= 736
Step= 832, Dmax= 1.1e-02 nm, Epot= -5.87424e+05 Fmax= 9.39117e+03, atom= 736
Step= 833, Dmax= 1.3e-02 nm, Epot= -5.87435e+05 Fmax= 1.00714e+04, atom= 736
Step= 835, Dmax= 7.6e-03 nm, Epot= -5.87495e+05 Fmax= 1.61871e+03, atom= 736
Step= 836, Dmax= 9.1e-03 nm, Epot= -5.87496e+05 Fmax= 1.23351e+04, atom= 736
Step= 837, Dmax= 1.1e-02 nm, Epot= -5.87575e+05 Fmax= 4.50901e+03, atom= 736
Step= 839, Dmax= 6.6e-03 nm, Epot= -5.87588e+05 Fmax= 5.57713e+03, atom= 736
Step= 840, Dmax= 7.9e-03 nm, Epot= -5.87602e+05 Fmax= 6.56343e+03, atom= 736
Step= 841, Dmax= 9.4e-03 nm, Epot= -5.87610e+05 Fmax= 7.96504e+03, atom= 736
Step= 842, Dmax= 1.1e-02 nm, Epot= -5.87617e+05 Fmax= 9.50139e+03, atom= 736
Step= 844, Dmax= 6.8e-03 nm, Epot= -5.87671e+05 Fmax= 9.67523e+02, atom= 736

writing lowest energy coordinates.

Steepest Descents converged to Fmax < 1000 in 845 steps
Potential Energy = -5.8767106e+05
Maximum force = 9.6752307e+02 on atom 736
Norm of force = 2.0982943e+01

GROMACS reminds you: "Physics is like sex: sure, it may give some practical results, but that's not why we do it" (Richard P. Feynman)
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ |

```

Fig 24: End of the energy minimisation process generating Potential Energy, Maximum Force and Norm of Force.

Note: Potential Energy (E_{pot}) should be negative, and for a simple protein in water on the order of 10^5 - 10^6 , depending on the system size and number of water molecules. This negative value indicates a stable and physically realistic conformation with favourable intermolecular interactions.

Maximum Force (F_{max}) is the largest force acting on any atom in the system during energy minimization in GROMACS. It represents the magnitude of the steepest force experienced by an atom due to interactions with neighbouring atoms, measured in $\text{kJ mol}^{-1} \text{nm}^{-1}$.

The goal of energy minimization is to reduce F_{max} below a predefined threshold (set in minim.mdp "emtol = 1000.0"), ensuring that all atoms are in a physically stable and low-energy conformation before running molecular dynamics simulations.

However, an issue arises when the Maximum Force (F_{max}) exceeds the convergence criterion set by emtol = 1000.0 (i.e., $1000 \text{ kJ mol}^{-1} \text{nm}^{-1}$), meaning the system has not fully relaxed into a stable minimum despite reaching an acceptable E_{pot} value. Evaluate why it may be happening, and perhaps change your minimization parameters (integrator, emstep, etc).

```

mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx energy -f em.edr -o potential.xvg
:-) GROMACS - gmx energy, 2025.0 (-:

Executable: /usr/local/gromacs/bin/gmx
Data prefix: /usr/local/gromacs
Working dir: /home/mansea/Gromacs_Weblem
Command line:
gmx energy -f em.edr -o potential.xvg

```

Fig 25: Analyse em.edr file using the GROMACS energy module.

```

Opened em.edr as single precision energy file

Select the terms you want from the following list by
selecting either (part of) the name or the number or a combination.
End your selection with an empty line or a zero.
-----
 1 Bond           2 Angle          3 Proper-Dih.    4 Ryckaert-Bell.
 5 LJ-14          6 Coulomb-14      7 LJ-(SR)        8 Coulomb-(SR)
 9 Coul.-recip.   10 Potential      11 Pressure      12 Vir-XX
13 Vir-XY         14 Vir-XZ         15 Vir-YX         16 Vir-YY
17 Vir-YZ         18 Vir-ZX         19 Vir-ZY         20 Vir-ZZ
21 Pres-XX        22 Pres-XY        23 Pres-XZ        24 Pres-YY
25 Pres-YY        26 Pres-YZ        27 Pres-ZX        28 Pres-ZY
29 Pres-ZZ        30 #Surf*SurfTen  31 T-rest
-----

10 0

Back Off! I just backed up potential.xvg to ./#potential.xvg.1#
Last energy frame read 668 time 844.000

Statistics over 845 steps [ 0.0000 through 844.0000 ps ], 1 data sets
All statistics are over 669 points (frames)

Energy                Average  Err.Est.      RMSD  Tot-Drift
-----
Potential              -565417    11000      28592  -74758.1 (kJ/mol)

GROMACS reminds you: "Alas, You're Welcome" (Prof. Dumbledore in Potter Puppet Pals)

```

Fig 26: Selecting "10 0" to select Potential (10); zero (0) terminates input; the average of E_{pot} shown and a file called "potential.xvg" will be written.

```

mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ xmgrace potential.xvg
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ |

```

Fig 27: Plotting the Potential Energy graph using xmgrace tool.

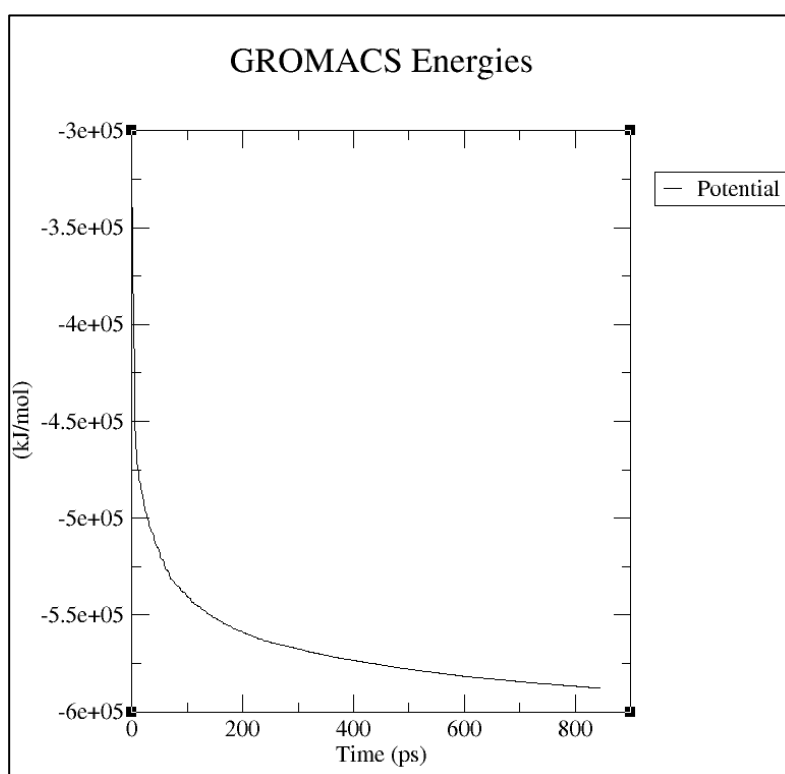


Fig 28: Potential Energy Graph; shows the potential energy (E_{pot}) decreasing over time, indicating that the system is stabilizing during the energy minimization process.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ gmx editconf -f em.gro -o em.pdb
:-) GROMACS - gmx editconf, 2025.0 (-:
```

Fig 29: Converting the em.gro file to em.pdb file.

```
Note that major changes are planned in future for trjconv, to improve usability and utility.
Will write pdb: Protein data bank file
Reading file em.tpr, VERSION 2021.4-Ubuntu-2021.4-2 (single precision)
Reading file em.tpr, VERSION 2021.4-Ubuntu-2021.4-2 (single precision)
Select group for output
Group 0 (      System) has 33876 elements
Group 1 (      Protein) has 1960 elements
Group 2 (      Protein-H) has 1001 elements
Group 3 (      C-alpha) has 129 elements
Group 4 (      Backbone) has 387 elements
Group 5 (      MainChain) has 517 elements
Group 6 (      MainChain+Cb) has 634 elements
Group 7 (      MainChain+H) has 646 elements
Group 8 (      SideChain) has 1314 elements
Group 9 (      SideChain-H) has 484 elements
Group 10 (      Prot-Masses) has 1960 elements
Group 11 (      non-Protein) has 31916 elements
Group 12 (      Water) has 31908 elements
Group 13 (      SOL) has 31908 elements
Group 14 (      non-Water) has 1968 elements
Group 15 (      Ion) has 8 elements
Group 16 (      CL) has 8 elements
Group 17 (      Water_and_ions) has 31916 elements
Select a group: 1
Selected 1: 'Protein'
Reading frame      0 time      0.000
```

Fig 30: Extracting the protein molecule from the solvation box for molecular docking; select group 1 (Protein).

Note: em.pdb file contains the energy-minimized structure which includes protein, solvent (SOL), and ions (e.g., Na^+ , Cl^-). GROMACS command trjconv is used to extract only the protein (or any other desired component) from a GROMACS structure file while removing unwanted molecules.

The command calls the trajectory conversion tool in GROMACS which utilise the input run input file (em.tpr) from the energy minimization step (used for indexing) and input structure file (contains protein + solvent + ions) to clean it and generate an output PDB file after selecting only the protein.

Reasons the trjconv might fail is if em.tpr was not generated correctly or if your .tpr file was generated in an older GROMACS version and you're running trjconv in a newer version, compatibility issues might arise.

```
mansea@DESKTOP-GMRN5C6:~/Gromacs_Weblem$ cp em_clean.pdb /mnt/c/Users/Mansi\ Rawat/Downloads/
```

Fig 31: Copying the em_clean.pdb file in Downloads folder in Windows.

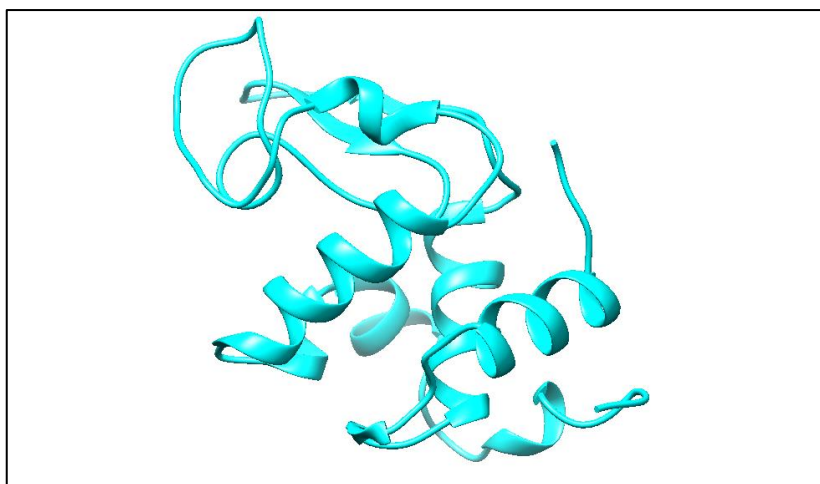


Fig 33: Visualising the structure em_clean.pdb in Chimera.

RESULTS:

The energy minimization of hen egg-white lysozyme (PDB ID: 1AKI) was successfully performed using GROMACS, following the standard workflow for molecular simulations. The potential energy (E_{pot}) graph obtained after minimization indicates a steady decrease in potential energy, confirming that the system reached a more stable and physically meaningful conformation. The initial potential energy was approximately -3.0×10^5 kJ/mol, which progressively declined to around -5.8×10^5 kJ/mol, demonstrating successful relaxation of the molecular structure. The final structure obtained from minimization was visualized using Chimera. Overall, the results validate that the energy minimization was effective, yielding a stable protein conformation suitable for further equilibration and production MD runs or molecular docking using appropriate ligands.

CONCLUSION:

Energy minimization in molecular simulations ensures physically realistic interactions and structural stability before proceeding to equilibration and production MD runs as well as a stable molecular docking. The energy minimization of PDB ID: 1AKI using GROMACS was successfully performed, ensuring that the molecular structure reached a stable, low-energy conformation before further molecular dynamics simulations. The stepwise approach included preprocessing the protein, solvation, ion addition, and minimization effectively reduced steric clashes and optimized the molecular geometry. The potential energy graph confirmed a steady decrease in energy, indicating convergence towards a physically stable structure, with the final potential energy reaching approximately -5.8×10^5 kJ/mol. The final minimized structure was visualized in Chimera, confirming the removal of steric clashes and the correctness of molecular geometry.

REFERENCES:

1. *About GROMACS* — GROMACS webpage <https://www.gromacs.org> documentation. (n.d.). <https://www.gromacs.org/about.html>
 2. Bank, R. P. D. (n.d.). *RCSB PDB - 1AKI: THE STRUCTURE OF THE ORTHORHOMBIC FORM OF HEN EGG-WHITE LYSOZYME AT 1.5 ANGSTROMS RESOLUTION*. <https://www.rcsb.org/structure/1AKI>
 3. Leach, A. R. (2001). *Molecular modelling: Principles and Applications*. Pearson Education.
 4. *Lysozyme in water*. (n.d.). <http://www.mdtutorials.com/gmx/lysozyme/index.html>
-

WEBLEM: 2 Introduction to Molecular Docking

AIM:

Introduction to Molecular Docking using PyRx.

INTRODUCTION:

Molecular docking is a pivotal computational technique extensively used in structure-based drug design (SBDD) to predict the interaction between small molecules (ligands) and biological macromolecules, typically proteins. This method simulates the binding process by determining the most favorable orientation, conformation, and interaction energy of a ligand within the active site of the target receptor. The docking process involves two critical components: a search algorithm, which explores possible binding poses, and a scoring function, which estimates the binding energy and stability of the ligand-receptor complex. Molecular docking serves as an efficient tool for virtual screening of large chemical libraries, allowing rapid identification of potential drug candidates. It plays a crucial role in early-stage drug discovery by reducing the time and cost associated with experimental screening. Additionally, it supports downstream applications such as lead optimization, pharmacophore modeling, and mechanistic studies. However, the reliability of docking outcomes depends on multiple factors, including receptor flexibility, solvation effects, and the precision of scoring functions used (Meng et al., 2011).

Overview of Molecular Docking

Molecular docking is a computational technique used to predict the preferred orientation of one molecule (typically a ligand) when bound to another (such as a protein). This method is instrumental in understanding molecular interactions and plays a crucial role in drug discovery. The docking process primarily involves two components:

1. **Search Algorithm:** This component explores possible conformations of the ligand within the receptor's binding site.
2. **Scoring Function:** This evaluates and ranks these conformations based on predicted binding affinity.

Types of Scoring Functions:

Scoring functions are essential in assessing the strength and stability of the ligand-receptor complex. They are generally categorized into:

- **Force-Field Based:** These functions sum intermolecular van der Waals and electrostatic interactions using molecular mechanics force fields like CHARMM or AMBER.
- **Empirical:** These functions count specific interaction types, such as hydrogen bonds or hydrophobic contacts, often parameterized using experimental data.

- **Knowledge-Based:** Derived from statistical observations of intermolecular contacts in large structural databases, these functions utilize statistical potentials to predict binding affinities.
- **Machine-Learning:** These functions employ data-driven approaches, using algorithms to predict binding affinities without assuming a predetermined functional form.

To streamline the molecular docking process, PyRx (Python Prescription) has emerged as an accessible and robust virtual screening software widely utilized in academic and preclinical research. PyRx integrates well-established docking engines such as AutoDock and AutoDock Vina, along with Open Babel for molecular file management and energy minimization. What sets PyRx apart is its intuitive graphical user interface (GUI), which simplifies complex docking workflows, making it user-friendly even for individuals with limited computational backgrounds. It allows researchers to conduct high-throughput molecular docking by offering integrated features for ligand import, energy minimization, receptor-ligand docking, and 3D pose visualization—all within a unified environment. Its cross-platform compatibility (Windows, Linux, macOS) and support for diverse molecular formats further enhance its utility. PyRx has been effectively applied in various studies for identifying lead compounds targeting viral proteins, enzymes, and receptors. For example, it has been employed in the screening of bioactive natural compounds against SARS-CoV-2 main protease and ACE2 receptor, as well as in evaluating phytochemicals for anticancer potential. Despite certain limitations—such as the assumption of receptor rigidity and the lack of solvent modeling—PyRx continues to serve as a foundational tool in virtual screening and hypothesis generation, often preceding more advanced computational or experimental validation steps (Dallakyan & Olson, 2015).

Key Features of PyRx

PyRx is an open-source virtual screening software widely used in structure-based drug discovery. Its intuitive interface and integrated tools make it suitable for academic research.

- **Docking Engines:** Supports AutoDock and AutoDock Vina, with Vina preferred for faster docking and better scoring accuracy (Aboelfetoh et al., 2021; Anand et al., 2022).
- **User-Friendly Interface:** A simple GUI allows users to perform docking without coding skills.
- **File Format Support:** Uses Open Babel to handle multiple formats like PDB, MOL, MOL2, SDF, PDBQT.
- **Ligand & Receptor Preparation:** Includes energy minimization (UFF), hydrogen addition, and format conversion tools.
- **Custom Grid Box:** Enables precise targeting of protein active sites.
- **High-throughput Screening:** Supports batch docking of multiple ligands.
- **Result Analysis:** Visualizes and exports binding affinities and poses for further analysis.
- **Cross-Platform & Free:** Available on Windows, Linux, macOS, and freely accessible for researchers.

Steps for PyRx Installation:

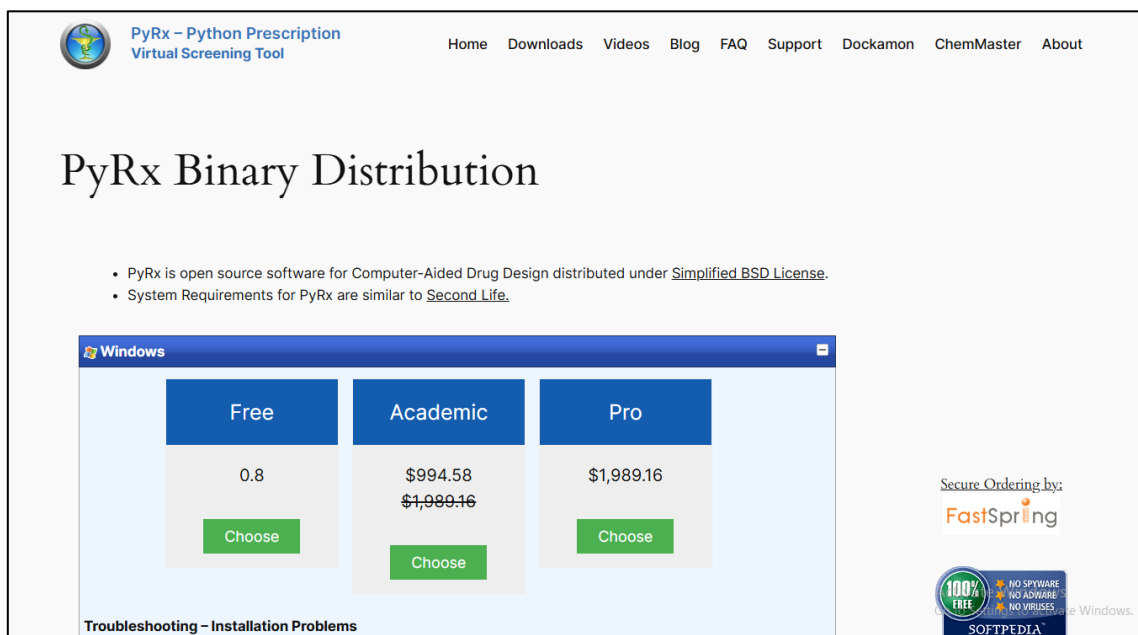


Fig 1: Webpage for PyRx Download

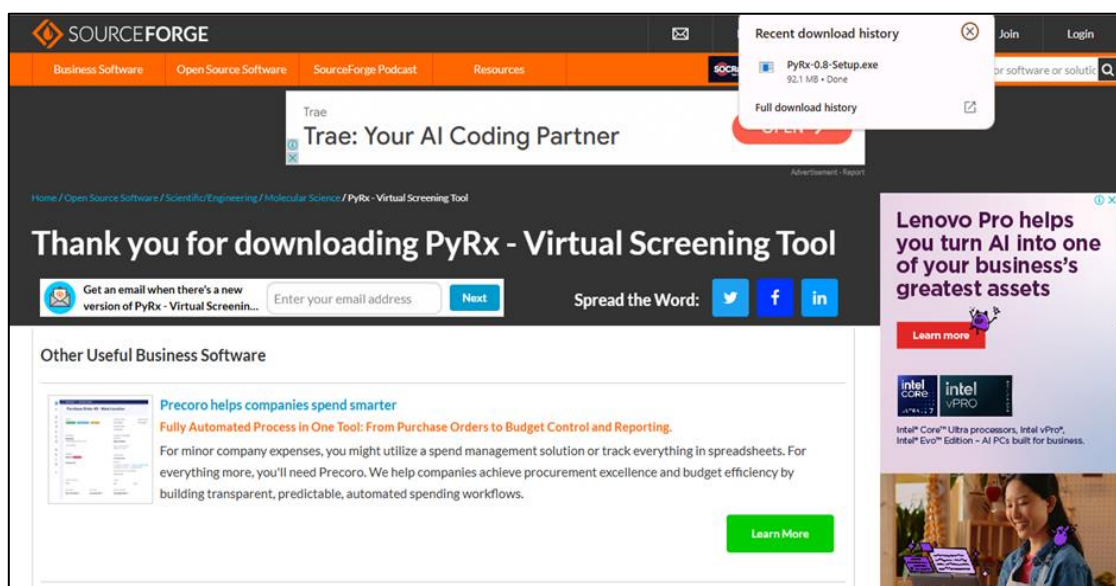


Fig 2: Downloading PyRx.Setup.exe

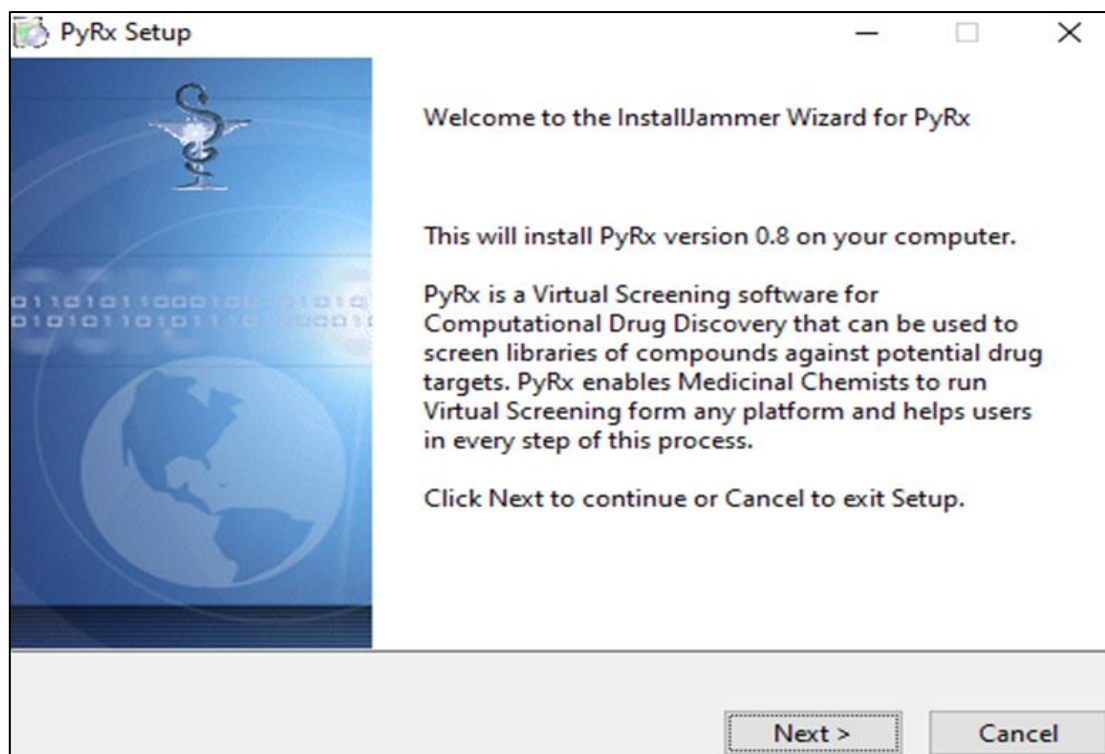


Fig 3: PyRx installation widget, click on next

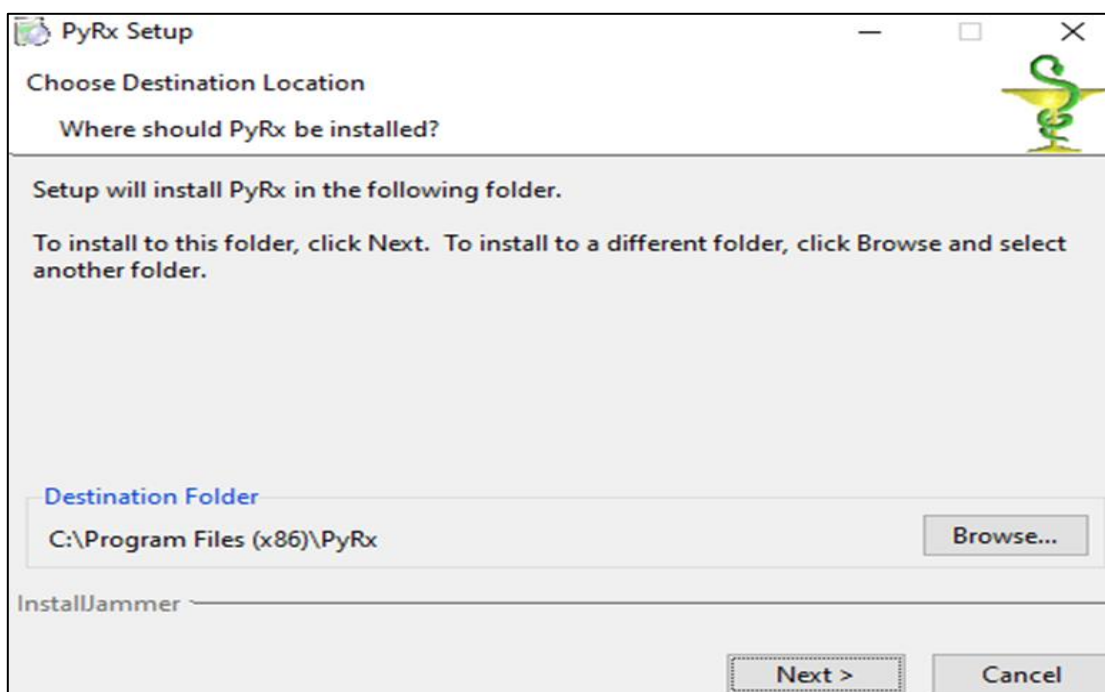


Fig 4: Giving the Path for installation

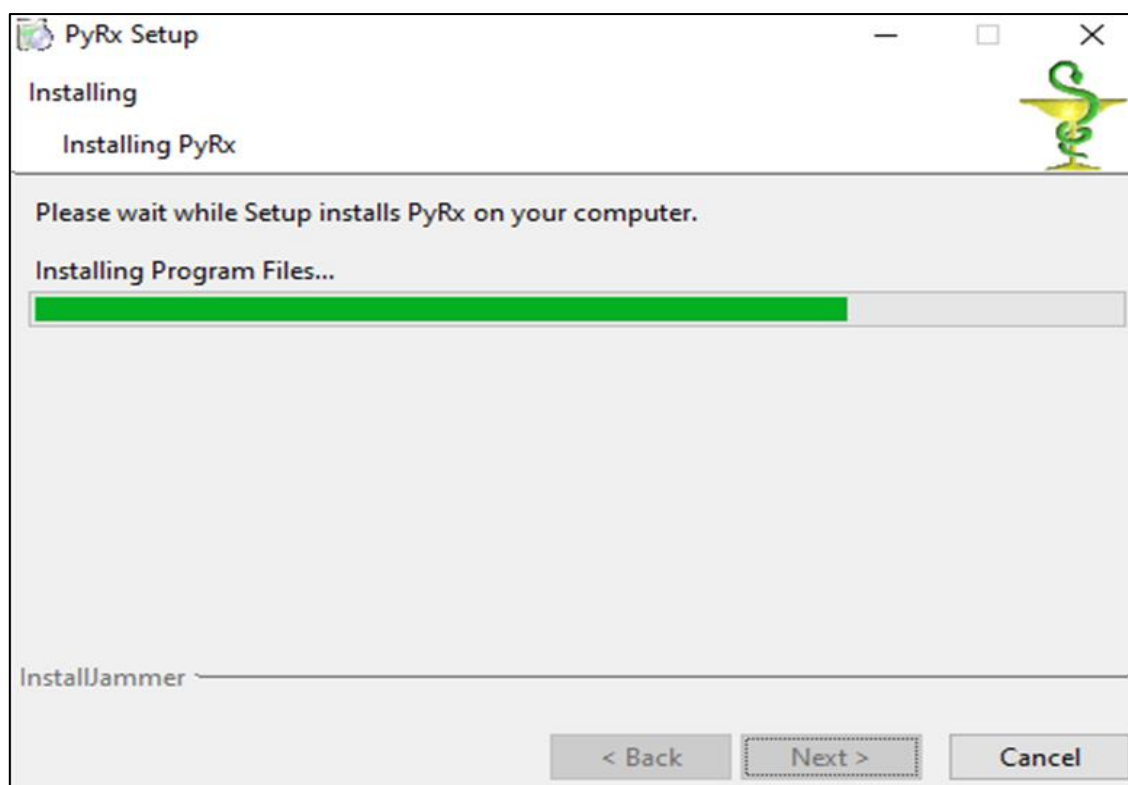


Fig 5: PyRx programme installing

REFERENCES:

1. Dallakyan, S., & Olson, A. J. (2015). Small-molecule library screening by docking with PyRx. *Methods in Molecular Biology*, 1263, 243–250. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4622462/>
2. Meng, X. Y., Zhang, H. X., Mezei, M., & Cui, M. (2011). Molecular docking: A powerful approach for structure-based drug discovery. *Current Computer-Aided Drug Design*, 7(2), 146–157. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3151162/>
3. Aboelfetoh, E. F., et al. (2021). *Journal of Biomolecular Structure and Dynamics*, 39(17), 6849–6859. <https://doi.org/10.1080/07391102.2021.1901734>
4. Anand, R., et al. (2022). *Journal of Biomolecular Structure and Dynamics*, 40(6), 2731–2740. <https://doi.org/10.1080/07391102.2021.1898082>

WEBLEM: 2(A)

PyRx- Python Prescription Virtual Screening Tool

(URL: <https://pyrx.sourceforge.io>)

AIM:

To perform Molecular docking for query “Lysozyme” (PDB ID: 1AKI) against a set of ligands (5 molecules) using PyRx Software.

INTRODUCTION:

Molecular docking is a common method employed in structure-based drug design. It is used in drug discovery to elucidate potential actions of uninvestigated bioactive compounds by identifying their molecular targets using bioinformatics along with systems biology approaches. It predicts how proteins interact with small chemical molecules or ligands to form a stable complex. In drug design and drug discovery the usefulness of molecular docking cannot be overemphasized. PyRx is an award-winning Virtual Screening software for Computational Drug Discovery that can be used to screen libraries of compounds against potential drug targets. PyRx enables Medicinal Chemists to run Virtual Screening from any platform and helps users in every step of this process – from data preparation to job submission and analysis of the results. While it is true that there is no magic button in the drug discovery process, PyRx includes docking wizard with an easy-to-use user interface which makes it a valuable tool for Computer-Aided Drug Design. PyRx also includes chemical spreadsheet-like functionality and powerful visualization engine that are essential for structure-based drug design.

PyRx is using a large body of established open source software including:

1. AutoDock 4 and AutoDock Vina are used as a docking software.
2. AutoDockTools, used to generate input files.
3. Python as a programming/scripting language.
4. wxPython for cross-platform GUI.
5. The Visualization ToolKit (VTK) by Kitware, Inc.
6. Enthought Tool Suite, including Traits, for application building blocks.
7. Open Babel for importing SDF files, removing salts and energy minimization.
8. matplotlib for 2D plotting.

Ligands, selected for docking are listed below:

Ligands	Name	pubchem (CID)
Standard	Metformin	4091
ligand 1	N-benzyl-N-methyl-1-phenylpropan-2-amine	2341
ligand 2	Pioglitazone	4829
ligand 3	Meridine	72473
ligand 4	Phenformin	8249

METHODOLOGY:

1. Prepare Input Files:

- a. Extract the protein structure from GROMACS simulation output in PDB format.
- b. Obtain the 3D structures of five ligands from PubChem in SDF format.

2. Launch PyRx and Load Files:

- Load Protein:
 - a. Click on 'File' → 'Load Molecule' and select your prepared protein file.
 - b. Once loaded, right-click on the protein in the workspace, select "AutoDock," → "Make Macromolecule." This converts the protein into PDBQT format.
- Load Ligands:
 - a. Open Bable , click on 'Insert new item' and Load ligands one by one in SDF format.
 - b. Minimize their energy by clicking on any of the cell and select 'Minimize All' to minimize the energy of all the ligands.
 - c. Convert ligands to PDBQT format by selecting "Convert All to AutoDock Ligand."

3. Set Up Docking Parameters:

- Open Vina Wizard:
 - a. Click on "Vina Wizard" to start setting up your docking experiment.
- Select Target and Ligands:
 - b. Select ligands and macromolecule from navigator → Autodock panel and use ctrl + shift buttons to select multiple ligands and press forward .
- Define Grid Box:
 - c. Adjust the grid box to encompass the active site of the target protein.
 - d. Use information from co-crystallized ligands or literature to determine the active site coordinates.

4. Run Docking:

1. After setting up the grid box, click "Forward" to proceed. Start the docking process by clicking "Run."
2. PyRx will now dock all ligands sequentially against the target protein. This may take some time depending on the number of ligands and system performance.

5. Analyze Results

1. Once docking is complete, view binding affinities for each ligand in the results table.
2. Identify ligands with the lowest binding energy (highest affinity). Save results by exporting them into CSV format for further analysis.
3. Save the ligands and protein file in .pdb format for further visualization.

OBSERVATIONS:

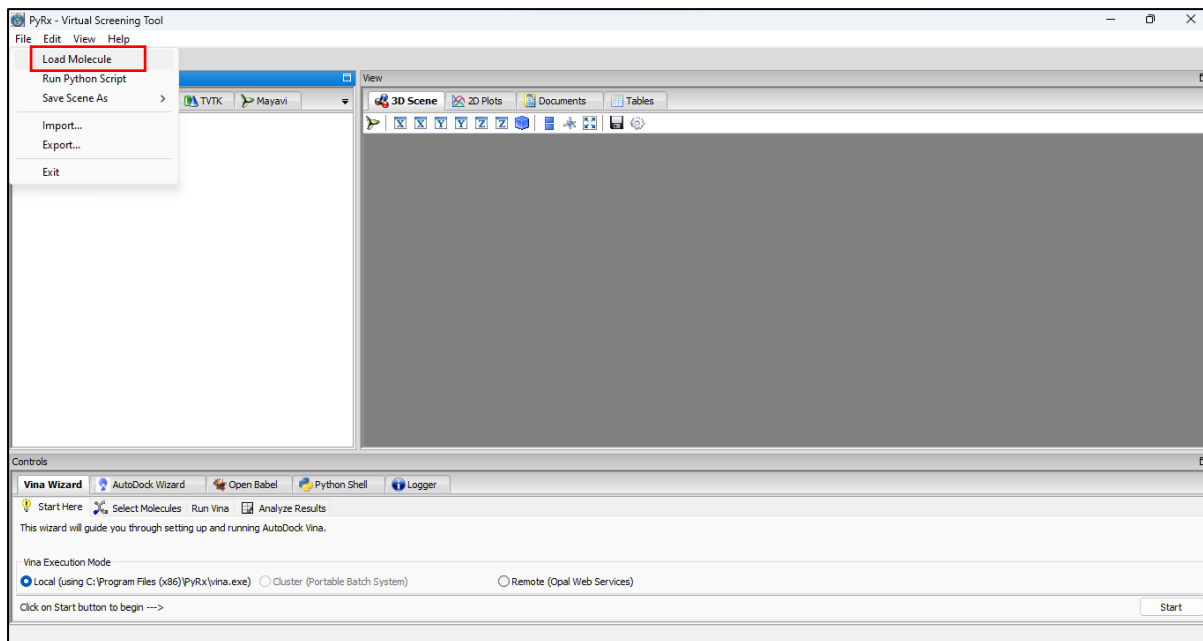


Fig 1: Navigate to the "File" menu and select "Load File" to initiate the process

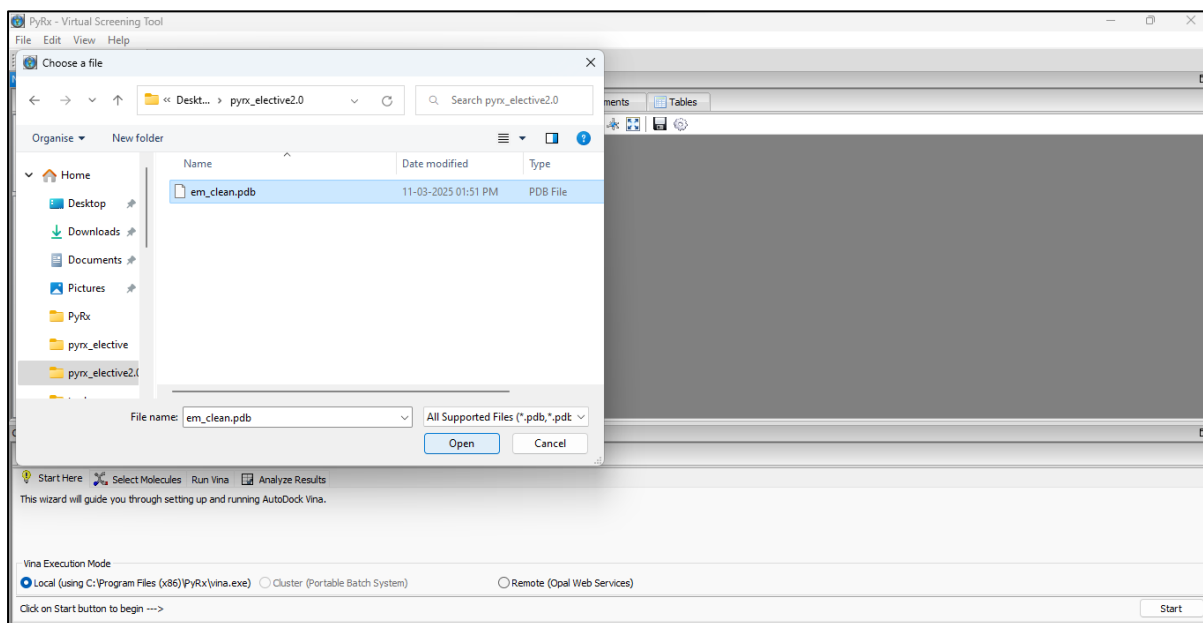


Fig 2: Loading the .pdb file generated from GROMACS simulations

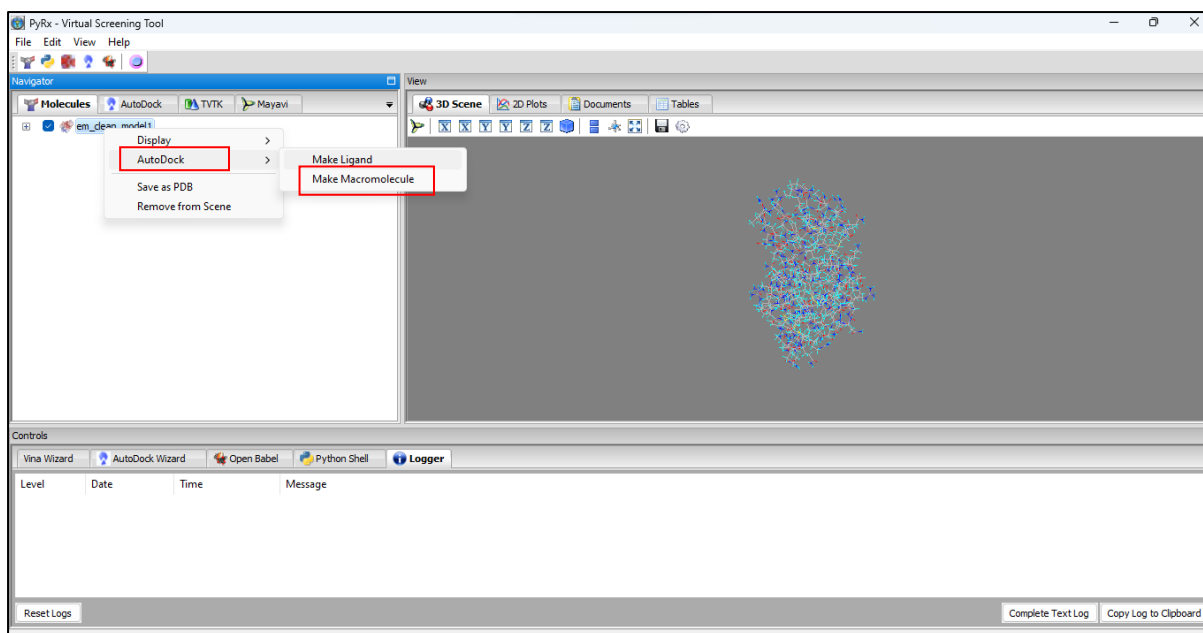


Fig 3: Converting the molecule into macromolecule

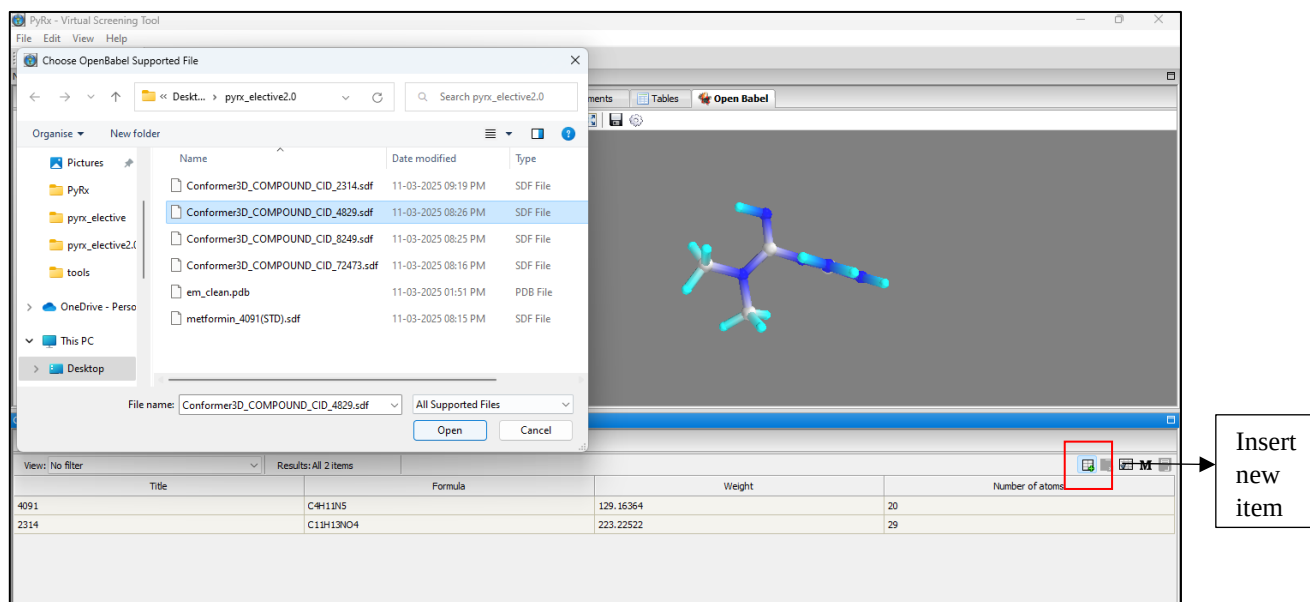


Fig 4: Insert ligands and Standard one by one

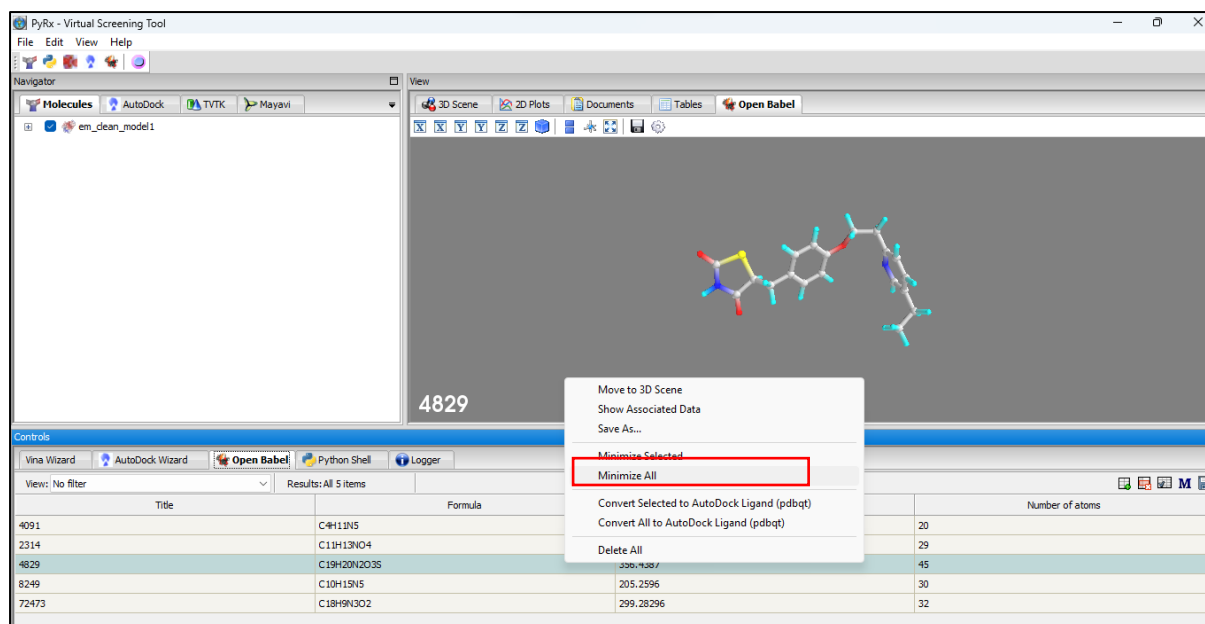


Fig 5: Minimize energy of all the ligands and standard

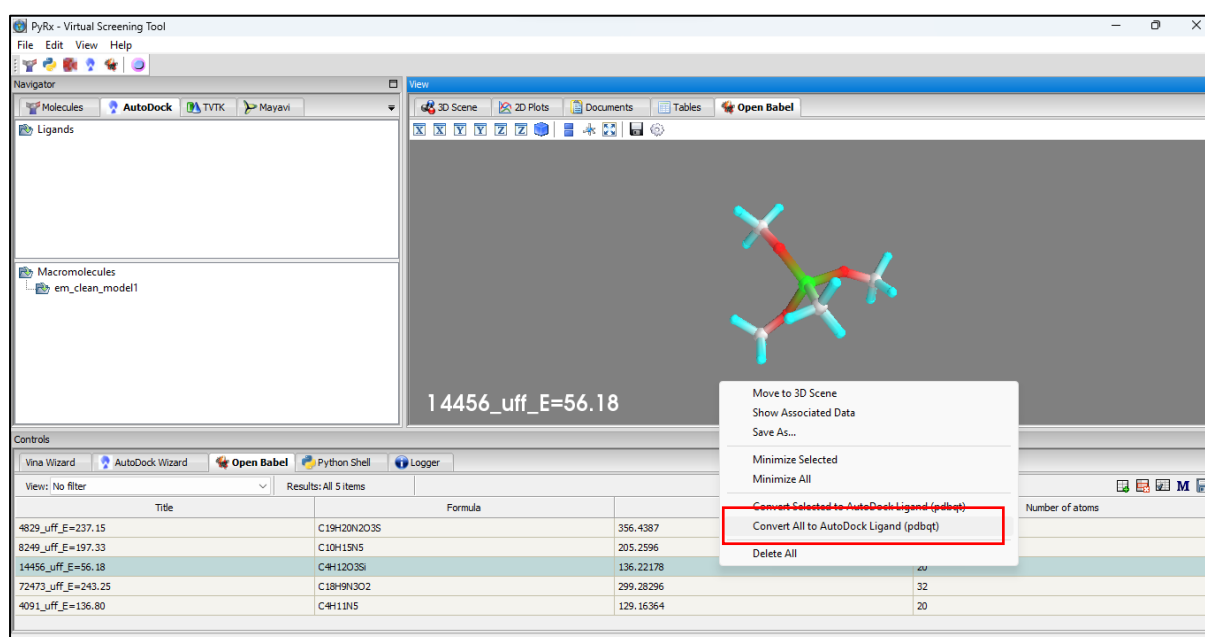


Fig 6: Converting all ligands and standard compound to AutoDock Ligand(pdbqt)

The screenshot shows the PyRx Virtual Screening Tool interface. The 'View' panel displays a table of ligand information. The 'Controls' panel at the bottom shows a list of items with columns for Title, Formula, Weight, and Number of atoms.

Name	Size	Date Created	Torsional DOF	AutoDock Elements
2314_uff_E=325.36	17	2025.03.11 21:3...	2	A C HD OA N
4091_uff_E=136.80	16	2025.03.11 21:3...	0	C HD N
4829_uff_E=237.15	27	2025.03.11 21:3...	7	A C OA N SA HD
72473_uff_E=243.25	26	2025.03.11 21:3...	0	A HD OA N
8249_uff_E=197.33	23	2025.03.11 21:3...	3	A C HD N

Title	Formula	Weight	Number of atoms
4091_uff_E=136.80	C ₉ H ₁₁ N ₅	129.16364	20
2314_uff_E=325.36	C ₁₃ H ₁₃ N ₄ O ₄	223.22522	29
4829_uff_E=237.15	C ₁₉ H ₂₀ N ₂ O ₃ S	356.4387	45
8249_uff_E=197.33	C ₁₀ H ₁₅ N ₅	205.2596	30
72473_uff_E=243.25	C ₁₈ H ₁₉ N ₃ O ₂	299.28296	32

Fig 7: Information about ligands generated on view panel

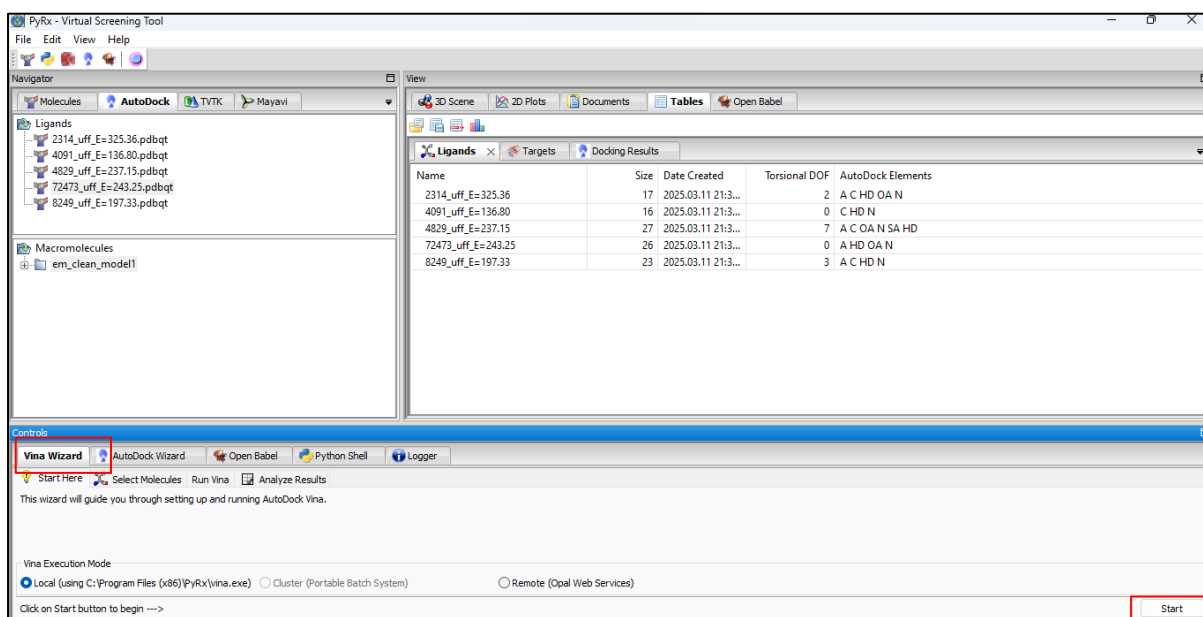


Fig 8: Click 'Vina Wizard' and then 'Start' for AutoDock

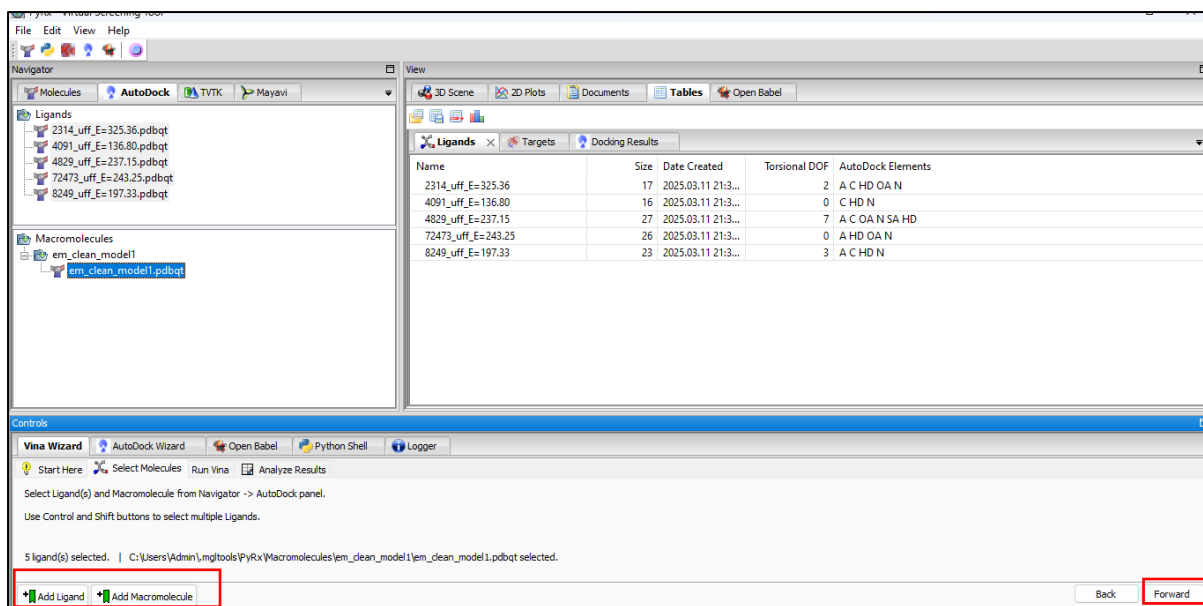


Fig 9: Add ligands and Macromolecules from navigator → AutoDock after that press forward

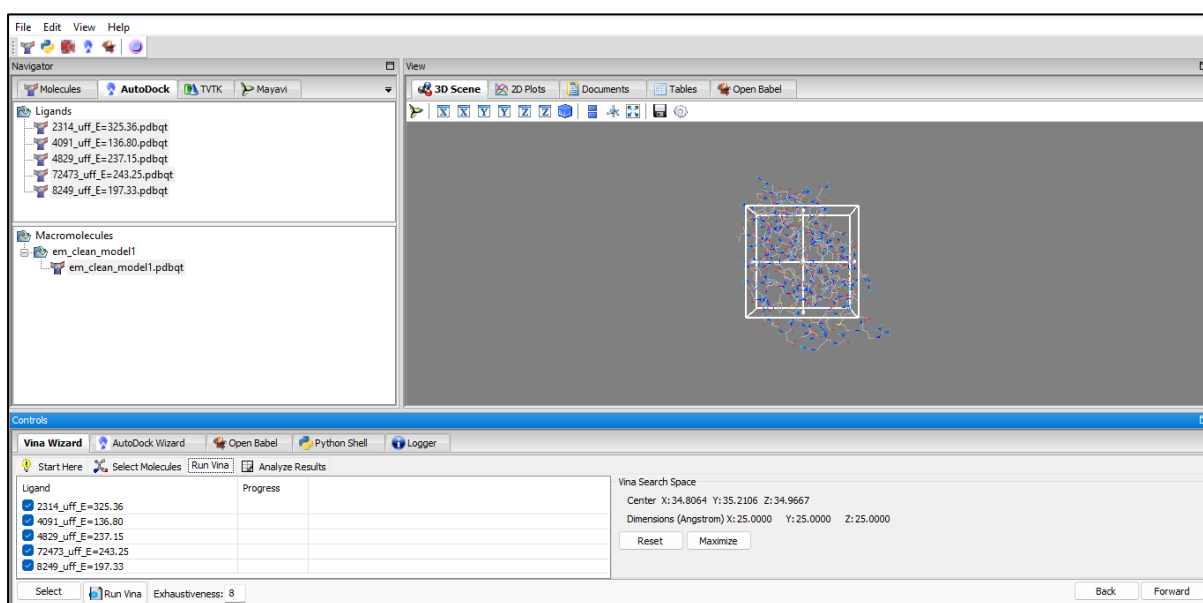


Fig 10: Adjust the grid to keep the whole molecule inside the grid

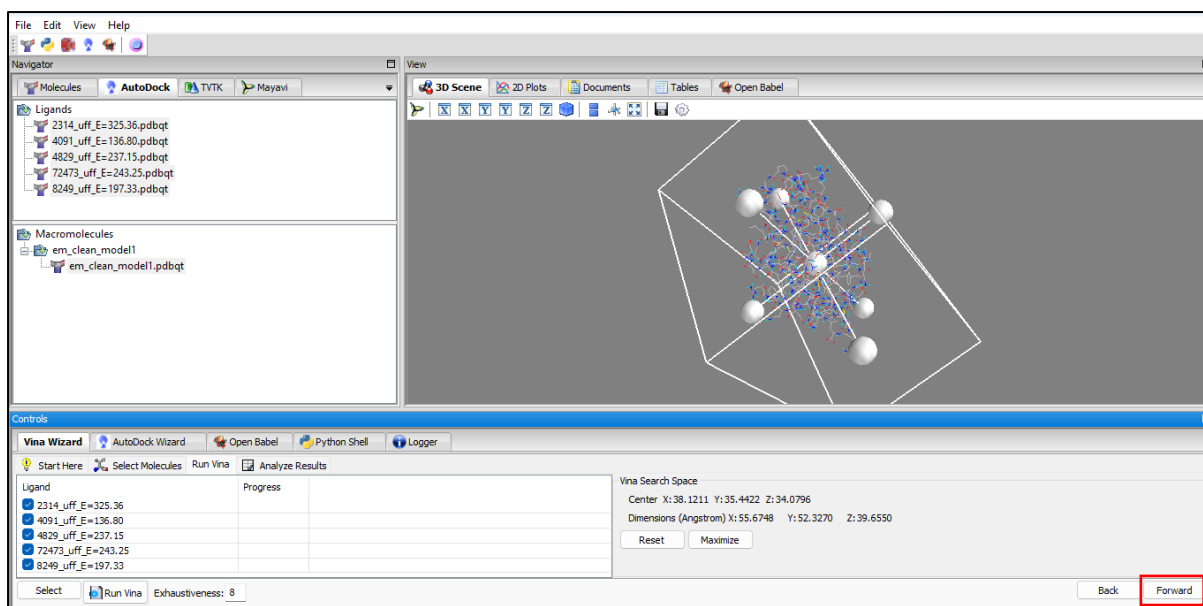


Fig 11: Adjusted Grid

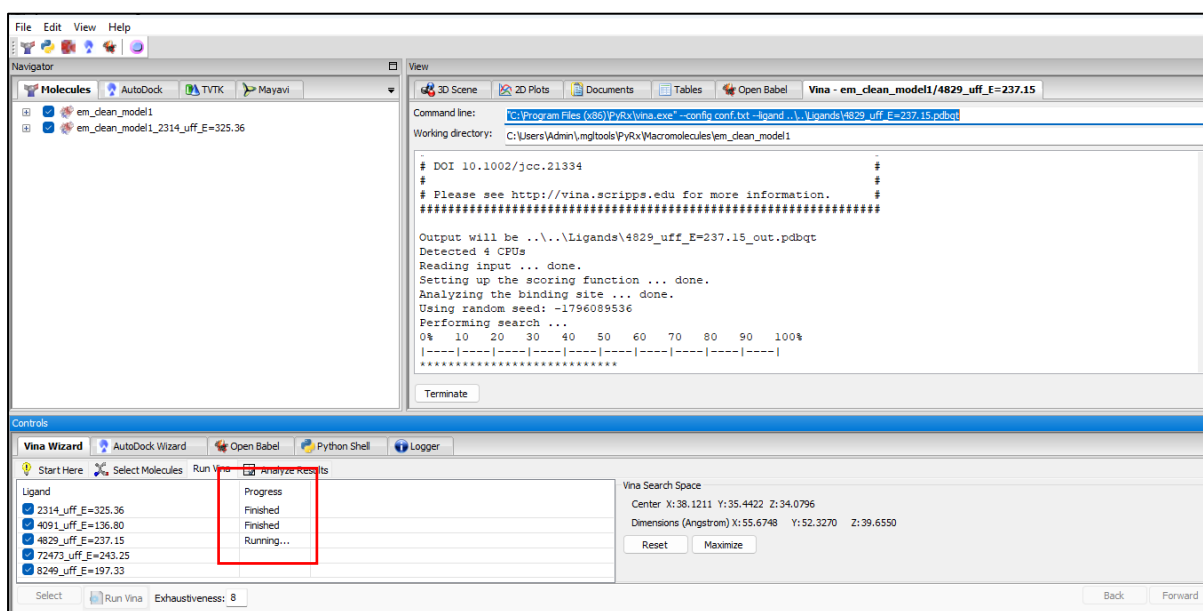


Fig 12: Analyzing Vina Search space

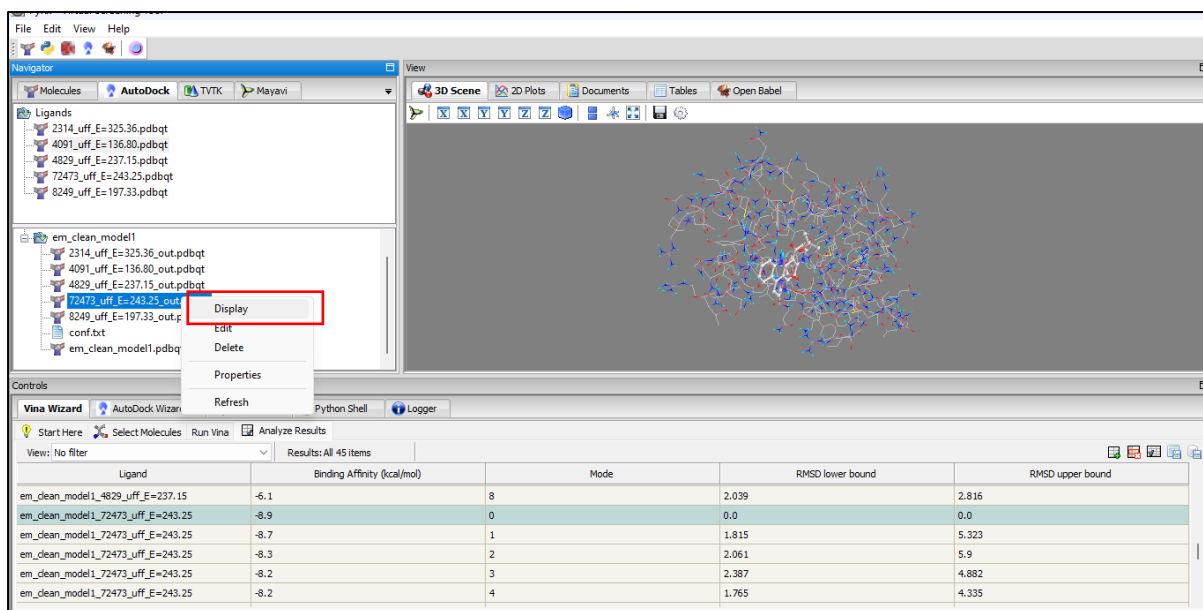


Fig 12: Display the selected Ligand

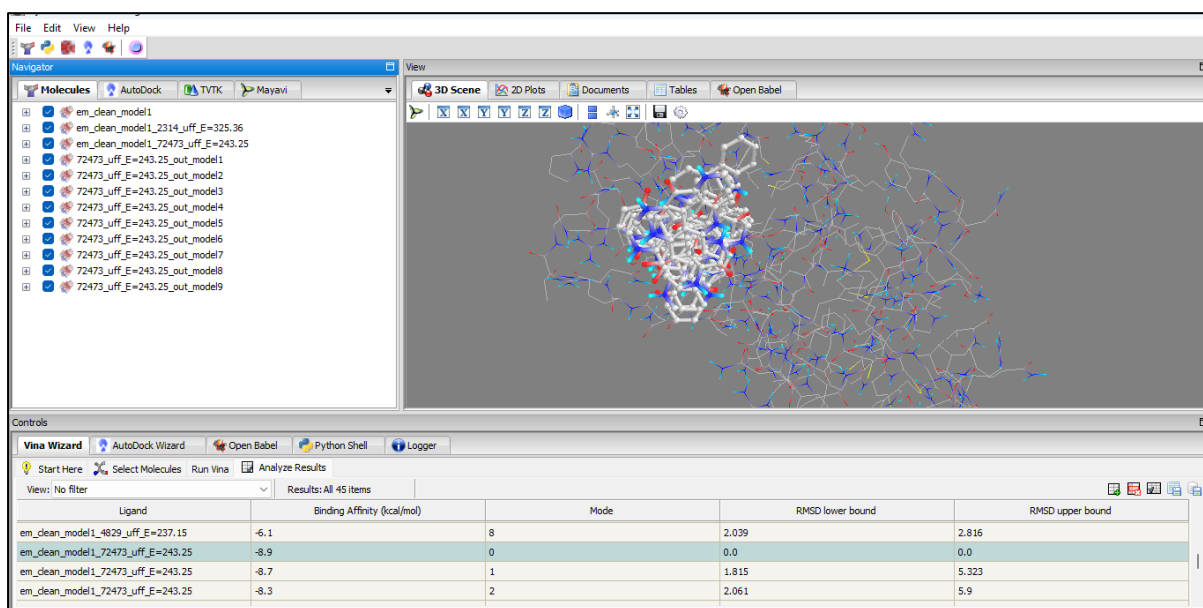


Fig 13: All the 9 conformers of each ligands can be displayed on the molecule

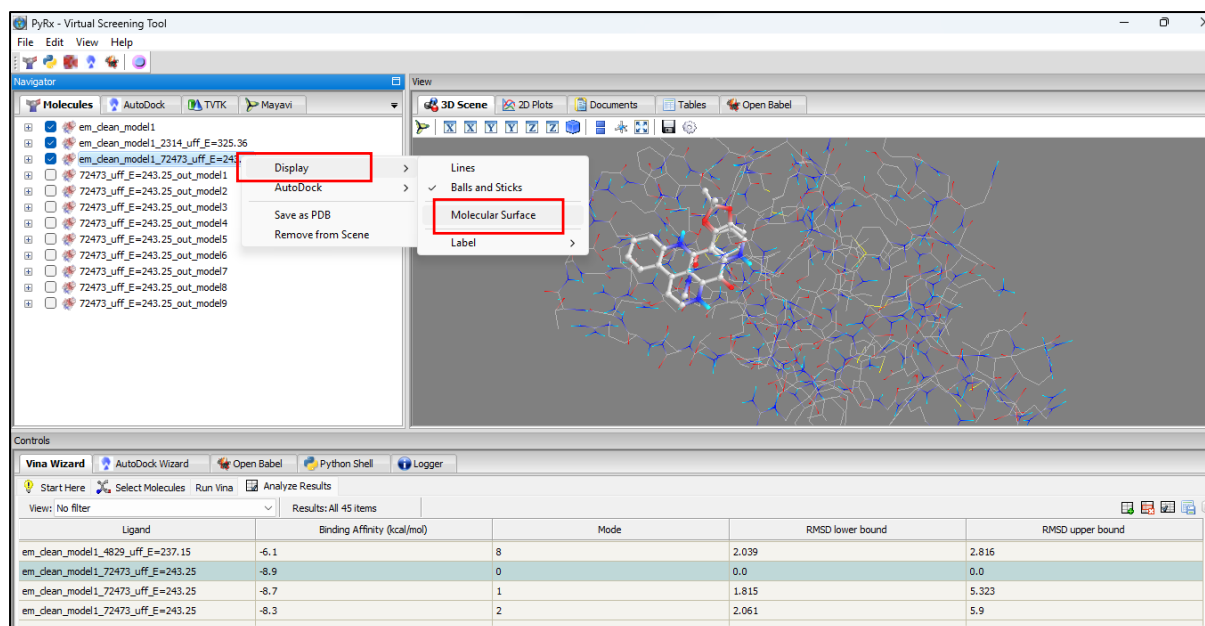


Fig 14: change the display to molecular surface

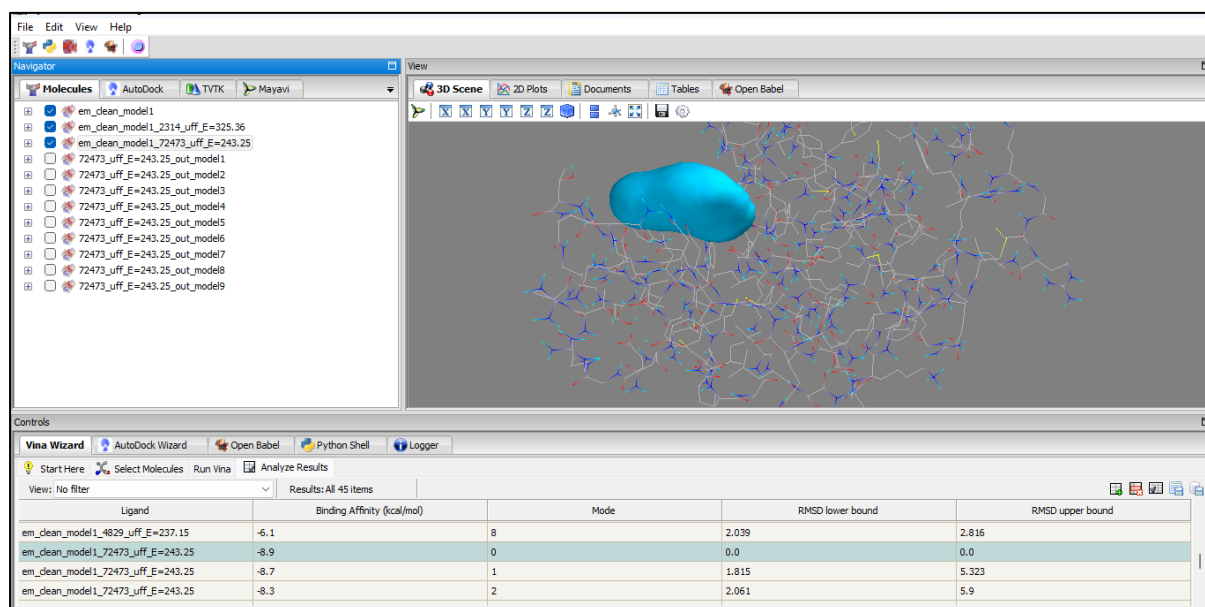


Fig 15: Changed Molecular surface

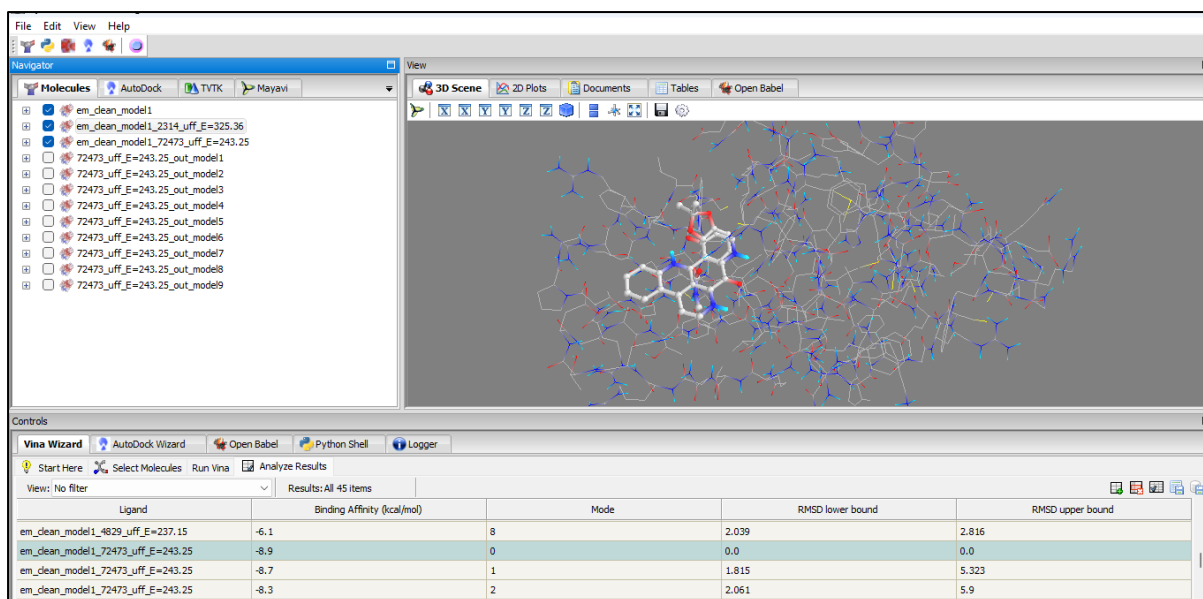


Fig 16: Select the Protein file and ligand with lowest binding affinity

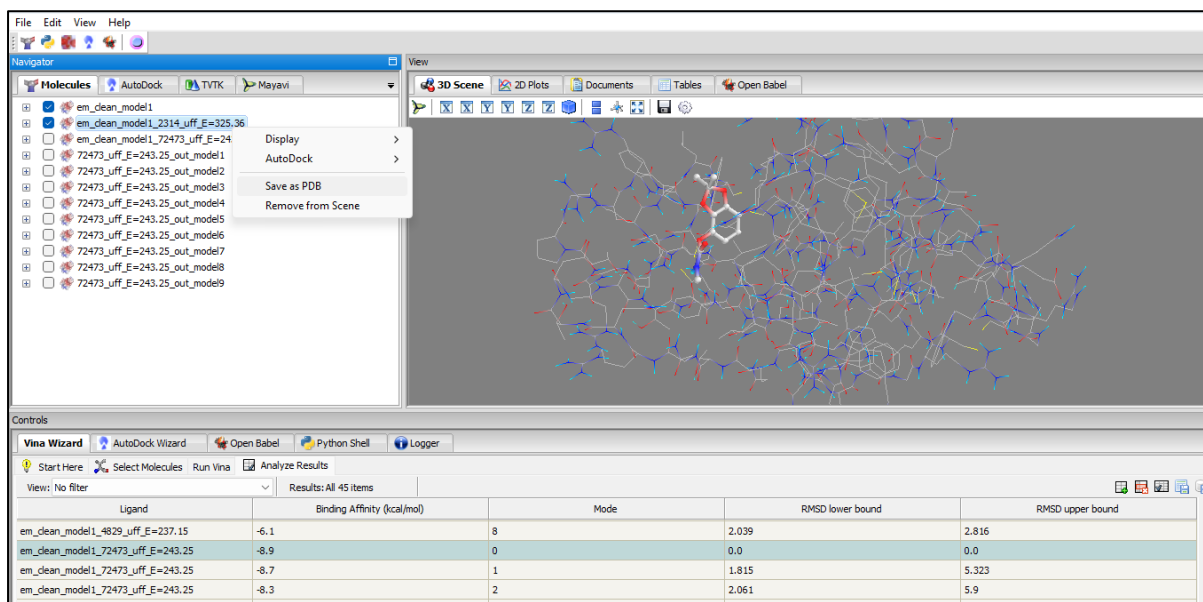


Fig 17: Save em_clean as pdb

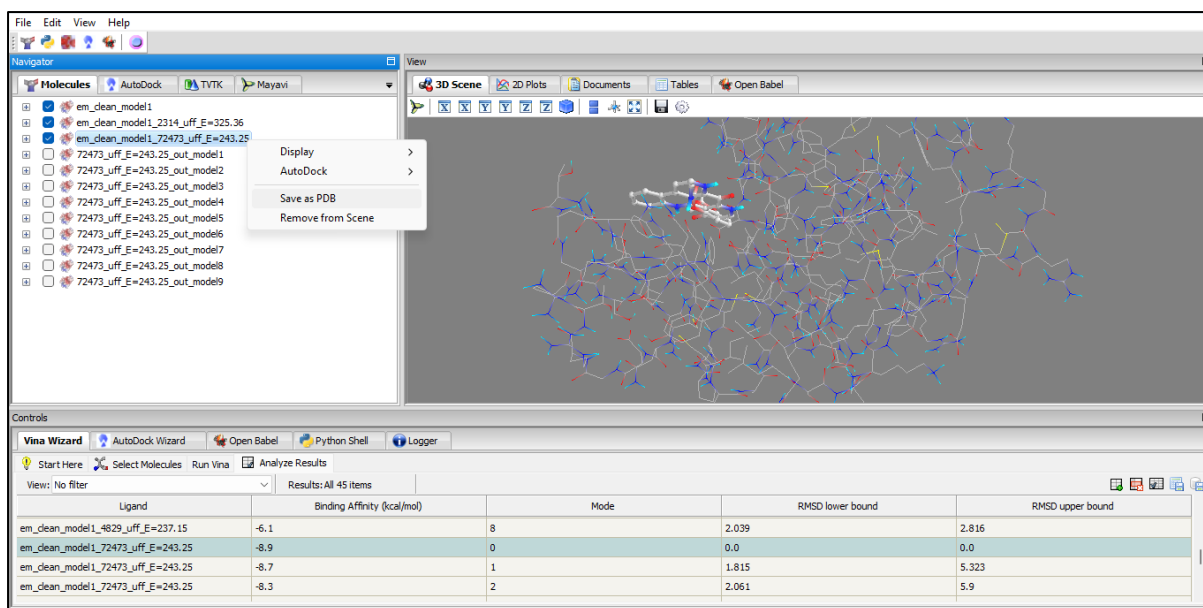


Fig 18: Save ligand as pdb

RESULTS:

Ligand	Binding Affinity	rmsd/ub	rmsd/lb
em_clean_model1_2314_uff_E=325.36	-6.2	0	0
em_clean_model1_4091_uff_E=136.80	-4.8	0	0
em_clean_model1_4829_uff_E=237.15	-6.8	0	0
em_clean_model1_72473_uff_E=243.25	-8.9	0	0

Standard Molecule: Metformin (CID: 4091) -- -4.8 (Binding Affinity)

Lowest Binding affinity molecule: Meridine (CID: 72473) -- -8.9 (Binding affinity)

Metformin, the standard molecule, exhibited a binding affinity of -4.8 kcal/mol, indicating a moderate interaction with the target protein. In contrast, Meridine demonstrated a significantly higher binding affinity of -8.9 kcal/mol, suggesting a more stable and potent interaction. This substantial difference implies that Meridine could be a more effective ligand than Metformin, potentially offering advantages in therapeutic applications. The superior binding affinity of Meridine makes it a promising candidate for further drug development or optimization, warranting additional investigation to validate its biological activity and explore its potential as a therapeutic agent.

CONCLUSION:

The utilization of PyRx software for molecular docking has provided valuable insights into the binding interactions between the target molecules and their respective ligands. Through this computational approach, we have identified potential candidates with promising binding affinities, shedding light on their suitability for further investigation in drug discovery and development processes. PyRx's user-friendly interface and robust algorithms have facilitated efficient virtual screening, allowing for the rapid evaluation of numerous compounds. The software's ability to analyze and visualize molecular interactions has enhanced our understanding of ligand-protein binding mechanisms, aiding in the identification of lead compounds for drug design endeavors.

REFERENCES:

1. PyRX – Python Prescription – Virtual Screening Tool. (n.d.).
<https://pyrx.sourceforge.io/>
 2. Admin, Admin, & Admin. (2024, October 8). A Comprehensive Guide to Virtual Screening with PyRx: Docking Multiple Ligands with a Macromolecule. Omics Tutorials - Bioinformatics, AI, Genomics, Proteomics and Transcriptomics.
<https://omicstutorials.com/a-comprehensive-guide-to-virtual-screening-with-pyrx-docking-multiple-ligands-with-a-macromolecule/>
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WEBLEM: 3

Introduction to Visualizing software

AIM:

Introduction to Discovery Studio Visualizer

INTRODUCTION:

Discovery Studio is an advanced, comprehensive suite of molecular modeling and simulation software developed by BIOVIA, designed to support life sciences researchers in the fields of computational chemistry, molecular modeling, bioinformatics, and structural biology. It offers a broad spectrum of tools that facilitate the visualization, modeling, simulation, and analysis of biological and chemical data. Whether for academic research, drug discovery, or industrial applications, Discovery Studio serves as a versatile platform that accelerates scientific discovery and innovation.

The core of Discovery Studio revolves around Discovery Studio Visualizer, a freely available application that allows users to visualize and interact with complex molecular structures, biological sequences, simulation results, and experimental data. The Visualizer supports a wide variety of data types and file formats, providing researchers with a cohesive environment for examining 3D molecular models, X-ray diffraction data, and molecular dynamics trajectories. It enables the preparation of publication-quality images and animations, fostering better communication of scientific findings.

Discovery Studio integrates a user-friendly graphical interface with powerful computational engines, enabling both novice users and expert computational scientists to perform intricate molecular tasks. These tasks range from structure-based drug design and pharmacophore modeling to quantitative structure-activity relationship (QSAR) studies and predictive ADME/Tox analyses. Through its flexible workflow design tools and automated protocols, Discovery Studio simplifies complex simulations and analyses that would otherwise require extensive programming knowledge.

The application runs on both Windows and Linux operating systems, offering a fully integrated desktop environment that utilizes standard OS features, including file system browsing, clipboard operations, and printing services. This ensures seamless access to data and resources throughout the modeling process.

Key Features of Discovery Studio Visualizer

The **Discovery Studio Visualizer** provides a rich set of interactive tools and viewers to support detailed molecular visualization and data analysis:

Feature	Description
Feature Search Box	A convenient search utility that helps users quickly locate and launch specific tools or functionalities by entering relevant keywords.
Menu Bar	Provides easy access to commonly used tools and commands for file handling, editing, visualization, analysis, and simulation tasks.
Toolbars	Contains shortcut icons for frequently used operations, improving workflow efficiency.
Tool Set Bar	Offers categorized tool panels such as "Pharmacophores" and "Ligand Interactions" for focused analysis and modeling tasks.
Windows Support	Allows simultaneous manipulation of multiple data types through different window types: - Molecule Window - Sequence Window - Script Window.
Status Bar	Displays real-time feedback and status information about ongoing operations or analysis tasks.
Files Explorer	Facilitates seamless browsing of the file system for importing, exporting, and managing molecular files and project data.

Scope and Capabilities of Discovery Studio

Discovery Studio addresses a wide range of computational needs in chemistry and biology through its extensive application areas:

1. Molecular Simulations

- Perform Molecular Mechanics (MM), Molecular Dynamics (MD), and Quantum Mechanics (QM) simulations.
- Support for implicit/explicit solvent models, membrane simulations, and hybrid QM/MM calculations.
- Investigate molecular motion, conformational changes, and thermodynamic properties.

2. Ligand Design

- Molecular library enumeration, virtual screening, and lead optimization.
- Build, refine, and evaluate chemical libraries for drug discovery campaigns.

3. Pharmacophore Modeling

- Generate 3D pharmacophore models for structure-activity relationship studies.
- Validate pharmacophore hypotheses and conduct virtual screening of compound libraries.

4. Structure-Based Drug Design (SBDD)

- Conduct receptor-ligand docking, fragment-based drug design, and de novo design of novel bioactive compounds.
- Analyze binding modes and predict binding affinities.

5. Macromolecule Modeling and Validation

- Tools for protein-protein docking, antibody modeling, and membrane protein design.
- Specialized features for G-protein coupled receptors (GPCRs) and antibody-antigen complexes.

6. Quantitative Structure-Activity Relationship (QSAR)

- Build predictive models using methods like multiple linear regression (MLR), partial least squares (PLS), and recursive partitioning.
- Perform 3D-QSAR analyses based on molecular fields.

7. ADME & Toxicity Predictions

- Predict pharmacokinetic properties like Absorption, Distribution, Metabolism, and Excretion (ADME).
- Evaluate the toxicity profiles of compounds to reduce risks in drug discovery.

User Experience and Workflow Efficiency

Discovery Studio's modular design and intuitive interface streamline research workflows:

- **Window Management:** Users can rearrange, dock, or float windows for a customized workspace.
- **Tool Panels:** Group tools by function for easy access to pharmacophore modeling, QSAR, or ligand interactions.
- **Workflow Automation:** Protocol-based workflows allow for reproducible and automated execution of complex analyses.

Installation:

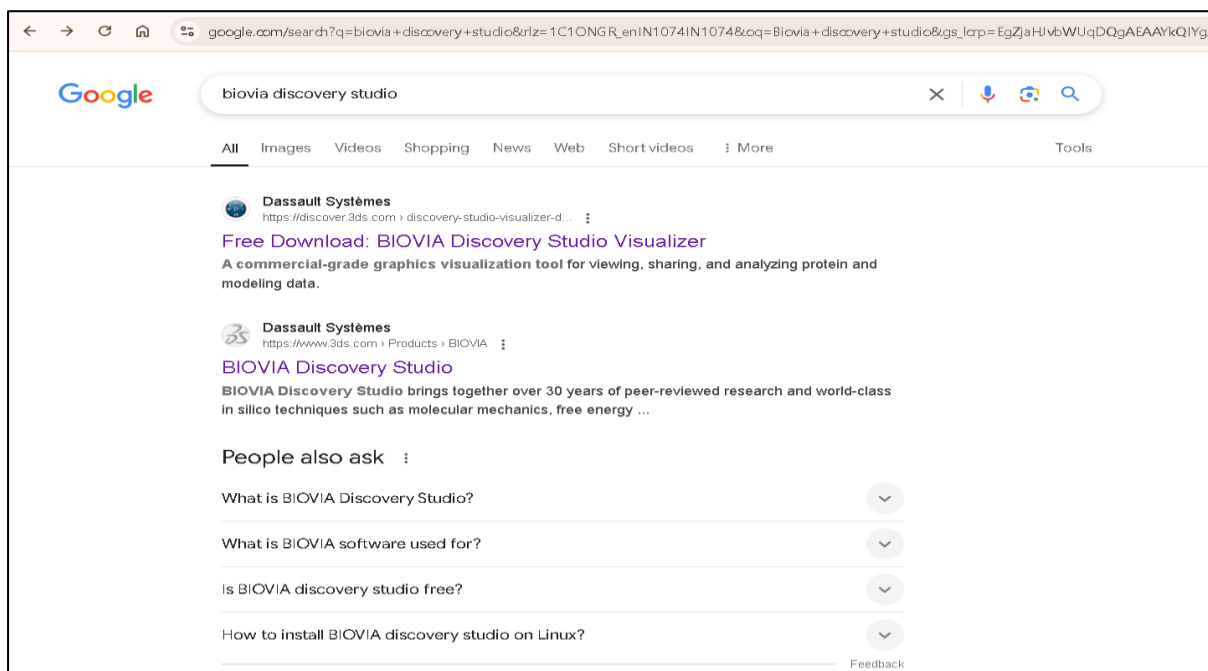


Fig.1: Google search results for "BIOVIA Discovery Studio." The first link directs users to the official Dassault Systèmes website for downloading the Discovery Studio Visualizer.

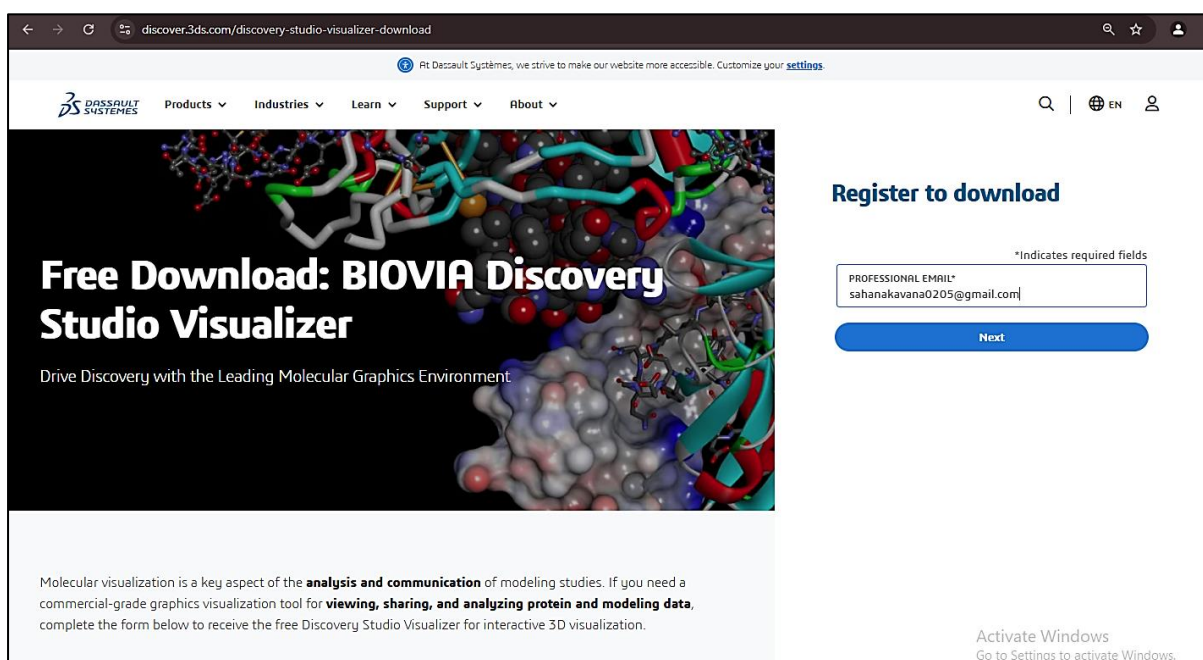


Fig.2: Official download page for BIOVIA Discovery Studio Visualizer on the Dassault Systèmes website. Users are required to register with a professional email address to access the download link.

Free Download: BIOVIA Discovery Studio Visualizer
Drive Discovery with the Leading Molecular Graphics Environment

Molecular visualization is a key aspect of the **analysis and communication** of modeling studies. If you need a commercial-grade graphics visualization tool for **viewing, sharing, and analyzing protein and modeling data**, complete the form below to receive the free Discovery Studio Visualizer for interactive 3D visualization.

Learn More about BIOVIA Discovery Studio

Register to download

Welcome, **sahanakavana0205@gmail.com**
[Click here to update](#)

*Indicates required fields

FIRST NAME*
Sahana

LAST NAME*
Poojary

COMPANY*
Gurunanak Khalsa College, Matunga

India

CITY*
Mumbai

Academic

Student

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Submit

Activate Windows
Go to Settings to activate Windows.

Fig.3: User registration form on the Dassault Systèmes BIOVIA Discovery Studio Visualizer download page. The form collects personal and professional details necessary to complete the registration for downloading the software.

Free Download: BIOVIA Discovery Studio Visualizer
Drive Discovery with the Leading Molecular Graphics Environment

Molecular visualization is a key aspect of the **analysis and communication** of modeling studies. If you need a commercial-grade graphics visualization tool for **viewing, sharing, and analyzing protein and modeling data**, complete the form below to receive the free Discovery Studio Visualizer for interactive 3D visualization.

Register to download

Welcome back, **Sahana Poojary**
Not you ? [Click here to update](#)

*Indicates required fields

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Submit

Activate Windows
Go to Settings to activate Windows.

Fig. 4: Final confirmation screen for registering on the Dassault Systèmes BIOVIA Discovery Studio Visualizer download page. The user agrees to the terms and conditions before submitting the registration form to access the download link.

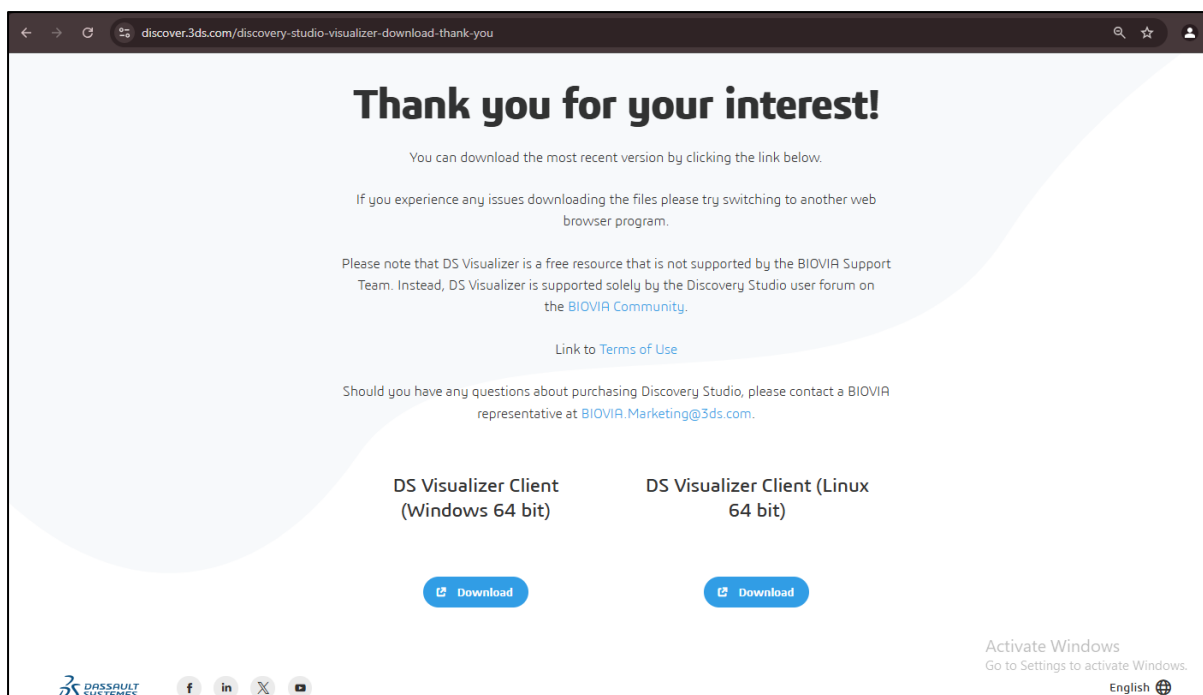


Fig.5: Download confirmation page for BIOVIA Discovery Studio Visualizer. The user can choose to download the DS Visualizer Client for either Windows 64-bit or Linux 64-bit systems after completing the registration process on the Dassault Systèmes website.

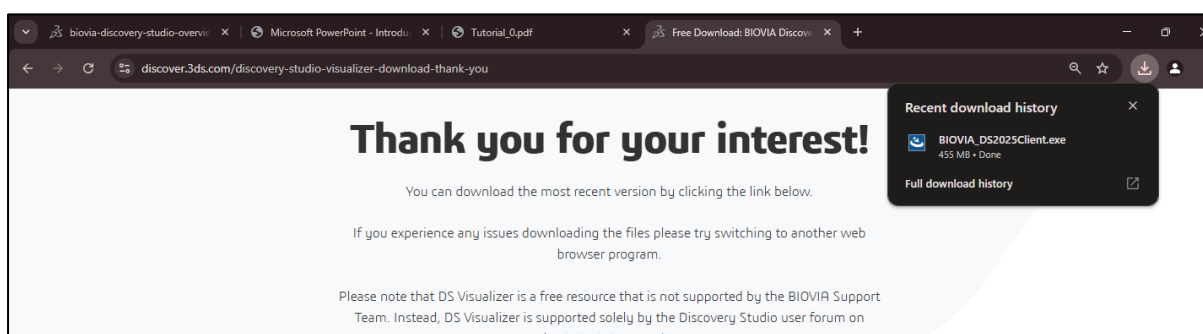


Fig.6: Successful download of the BIOVIA Discovery Studio Visualizer installer file (BIOVIA_DS2023Client.exe) as shown in the browser's recent download history popup, along with the final download confirmation page.

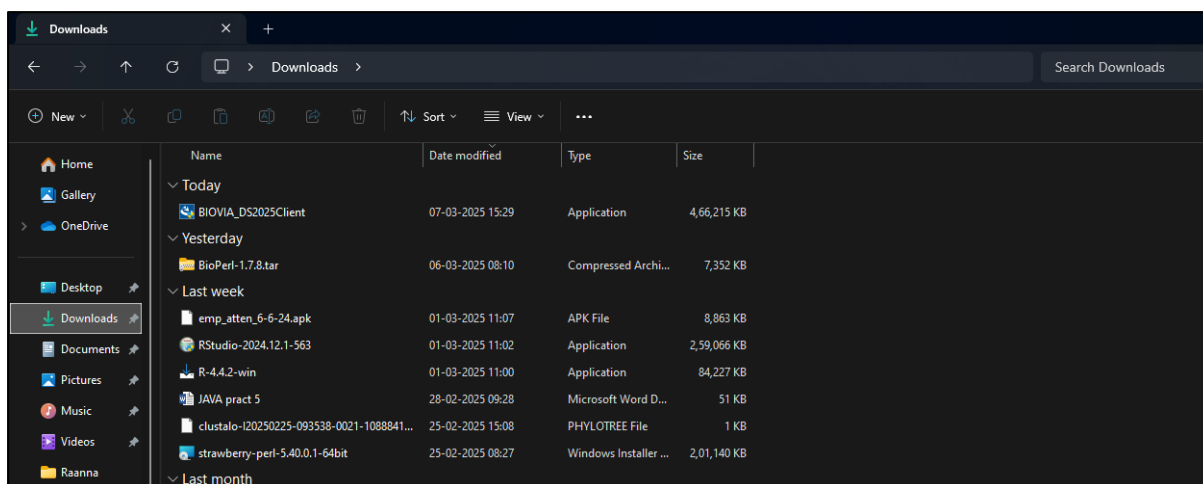


Fig.7: Downloads folder displaying various downloaded files, including BIOVIA Discovery Studio 2025 Client

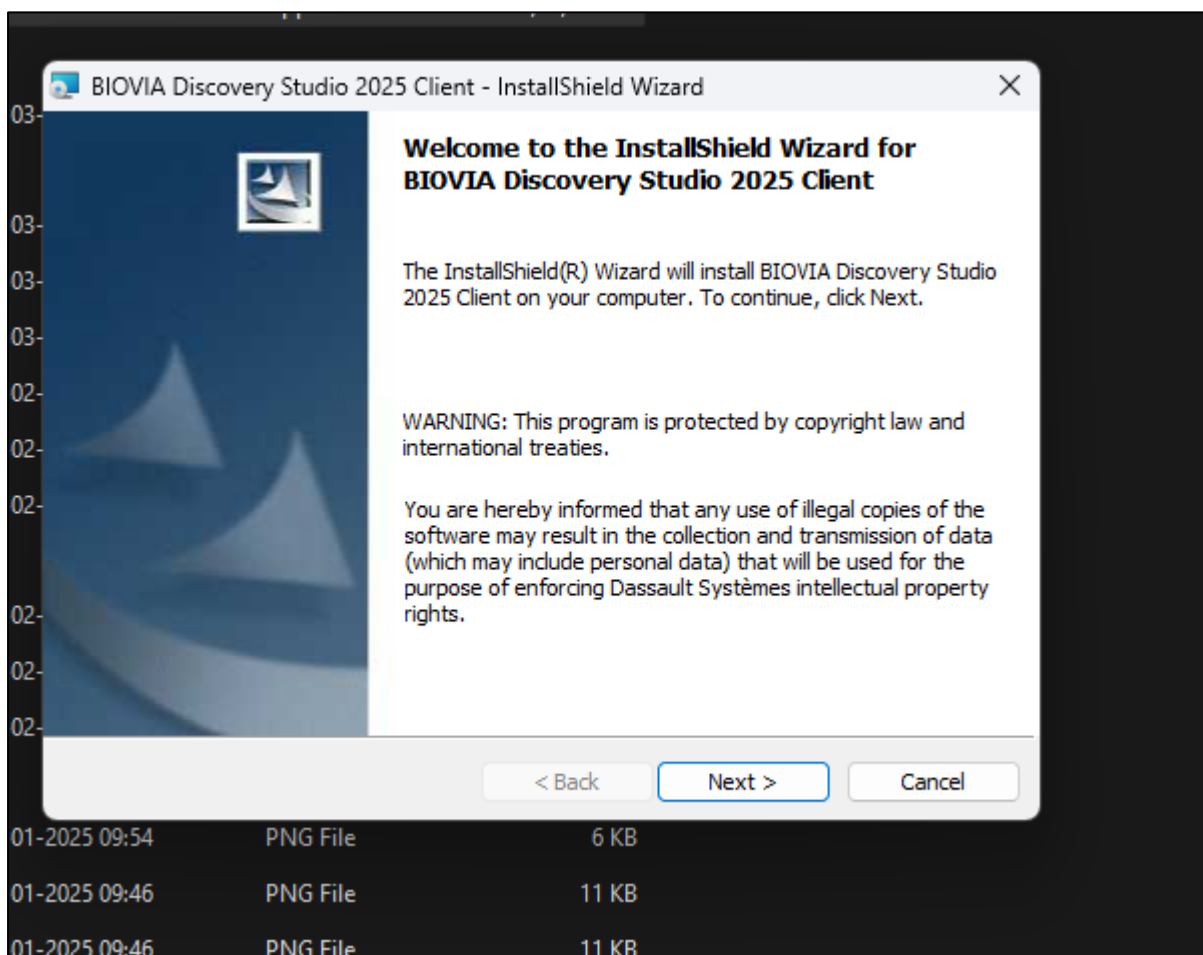


Fig.8: InstallShield Wizard welcome screen for BIOVIA Discovery Studio 2025 Client, displaying copyright information, warnings against illegal use, and instructions to proceed with the installation.

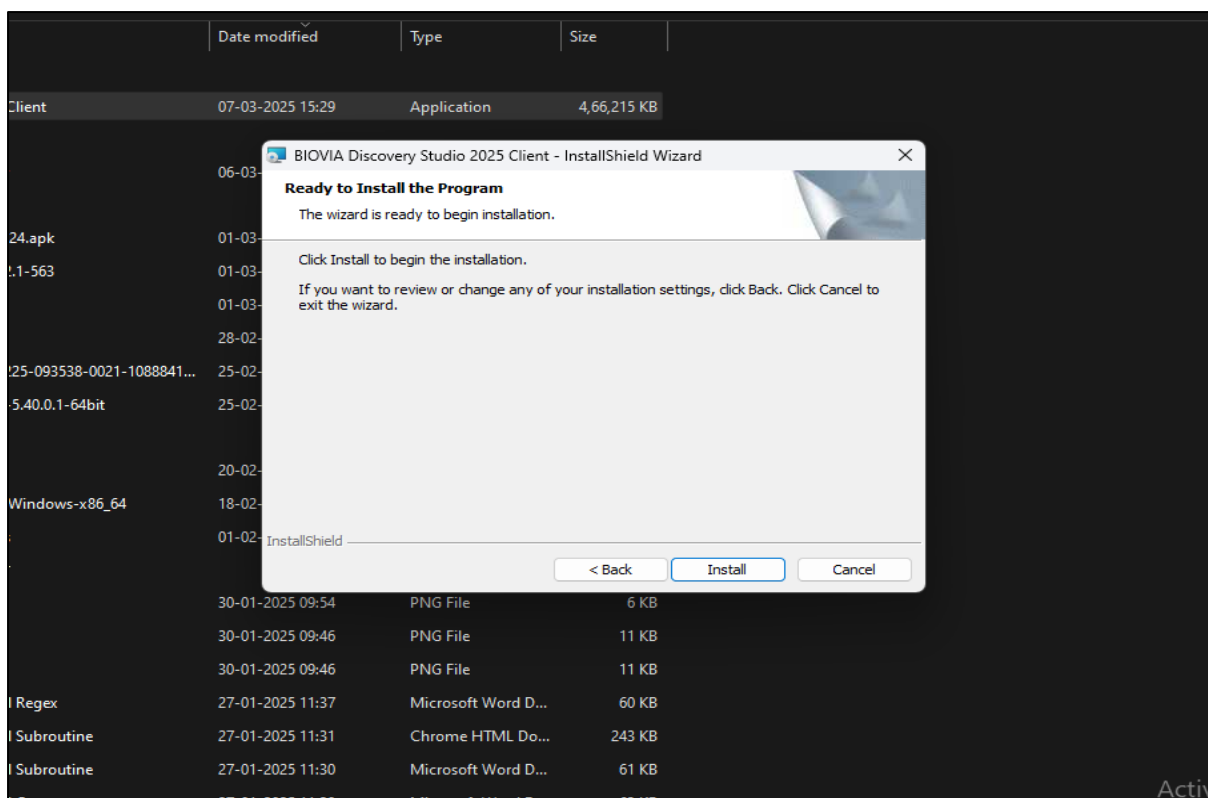


Fig.9: InstallShield Wizard confirmation screen for BIOVIA Discovery Studio 2025 Client, prompting the user to begin the installation process or review prior settings.

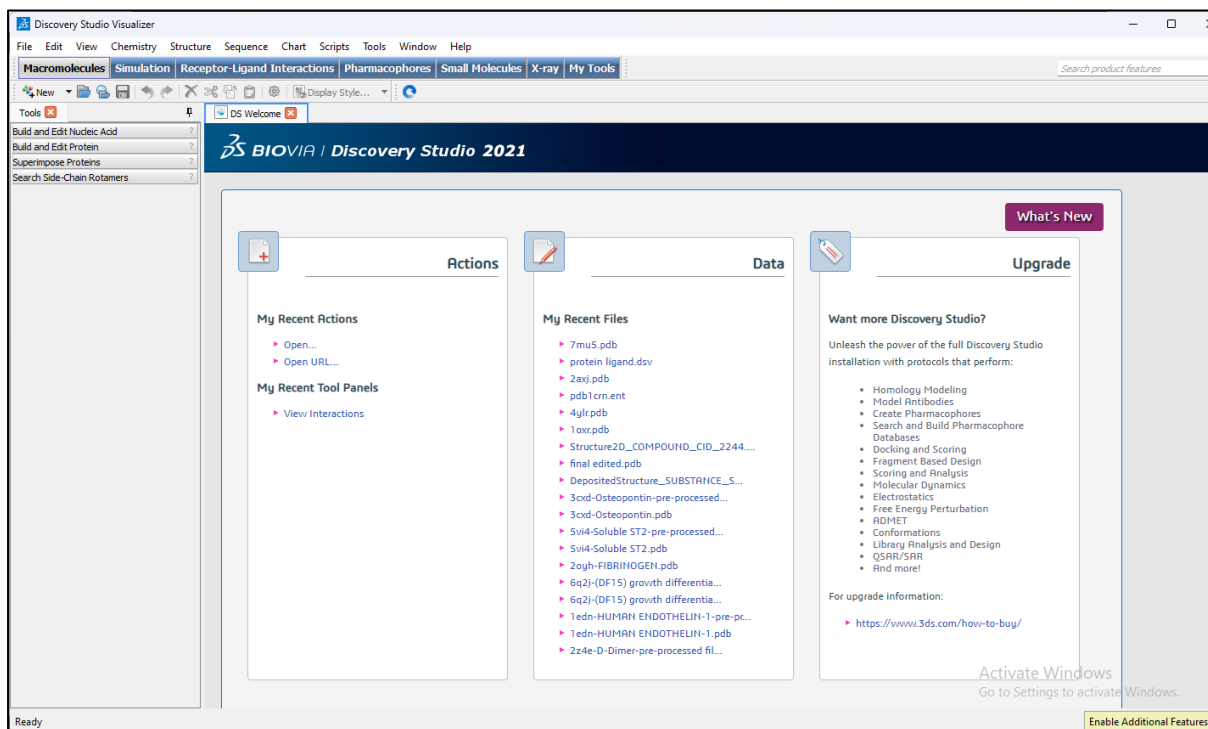


Fig.10: BIOVIA Discovery Studio Visualizer 2021 interface displaying recent actions, recent files, and upgrade options along with available tool panels for molecular modelling tasks

REFERENCES:

1. Martinelli, A. (n.d.). *Fondamenti - Tutorial 0*. Retrieved March 12, 2025, from http://www.adrianomartinelli.it/Fondamenti/Tutorial_0.pdf
 2. CUTM Courseware. (2020, September). *Introduction to Biovia Discovery Studio*. Retrieved March 12, 2025, from <http://courseware.cutm.ac.in/wp-content/uploads/2020/09/Introduction-to-Biovia-Discovery-Studio.pdf>
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WEBLEM: 3(A)

BIOVIA Discovery Studio Visualizer

AIM:

To perform discovery studio visualizer for protein structure Lysozyme (1AKB) and ligand Meridine (CID: 72473) retrieved from docking studies.

INTRODUCTION:

BIOVIA Discovery Studio is a comprehensive software suite designed for life sciences discovery research, particularly in the fields of drug discovery and biotherapeutic development. Developed by Dassault Systèmes BIOVIA, this platform integrates advanced in silico techniques with over 30 years of peer-reviewed research to provide a robust environment for molecular modeling and simulation. It offers a wide range of tools for molecular mechanics and dynamics, quantum mechanics, structure-based drug design, and ligand-based drug design, including receptor-ligand docking and pharmacophore modeling.

One of the key strengths of Discovery Studio is its ability to support a collaborative environment. It integrates seamlessly with BIOVIA Pipeline Pilot, allowing for workflow automation and integration with third-party tools. Additionally, the Discovery Studio Visualizer is a free tool that enables researchers to view, share, and analyze protein and small molecule data, facilitating collaboration among teams. This collaborative aspect is crucial for accelerating innovation in drug discovery and biotherapeutic development.

Discovery Studio also provides advanced tools for predicting absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties, which are essential for drug development. It supports methods like multiple linear regression and 3D field-based QSAR for designing therapeutics with favorable pharmacokinetic properties. Furthermore, it includes specialist tools for protein-protein docking, antibody design, and membrane-bound proteins like GPCRs, making it a versatile platform for both small and large molecule research.

Overall, BIOVIA Discovery Studio is a powerful tool for life sciences researchers, offering a comprehensive suite of validated science applications to accelerate innovation in drug discovery and biotherapeutic development. Its applications span from identifying and optimizing small and large molecule therapeutics to supporting the design and analysis of biotherapeutics. By enhancing productivity and collaboration, Discovery Studio plays a critical role in advancing life sciences research.

Meridine

Meridine (CID 72473) is a bisbenzylisoquinoline alkaloid known for its potential antimicrobial, anti-inflammatory, antioxidant, and antitumor properties. Structurally, it consists of two benzylisoquinoline units, contributing to its bioactivity. Similar alkaloids exhibit antibacterial and antifungal effects by interfering with microbial enzymes, while their anti-inflammatory properties stem from inhibiting pro-inflammatory mediators. Additionally, Meridine may act as an antioxidant, protecting cells from oxidative stress, and has shown potential anticancer activity by inducing apoptosis in tumor cells. Some

bisbenzylisoquinoline alkaloids also function as calcium channel blockers, suggesting cardiovascular benefits. Naturally found in medicinal plants like *Tinospora crispa* and *Stephania tetrandra*, these compounds have been traditionally used in Ayurvedic and Chinese medicine. Beyond its pharmacological effects, Meridine may play a role in neuroprotection and immune system modulation, making it a promising candidate for future drug development. However, its bioavailability, toxicity, and precise mechanisms of action remain largely unexplored, necessitating further research, in vitro studies, and clinical trials to assess its full therapeutic potential.

METHODOLOGY:

1. Open BIOVIA Discovery Studio.
2. Click on the File button, then click Open to load the molecules into Discovery Studio.
3. Load the energy-minimized protein file from the PyRex folder.
4. Click Open to load the ligand file, allowing it to be visualized alongside the protein structure.
5. Navigate to the View section and individually select Hierarchy for both the ligand and protein molecules.
6. Go to the Hierarchy section of the ligand molecule, right-click on it, and select Copy.
7. Paste the copied ligand into the Hierarchy section of the protein structure.
8. Click on 2D Diagram under 'Show receptor-ligand interaction in a 2D Diagram' to visualize the 2D interactions.

OBSERVATION:

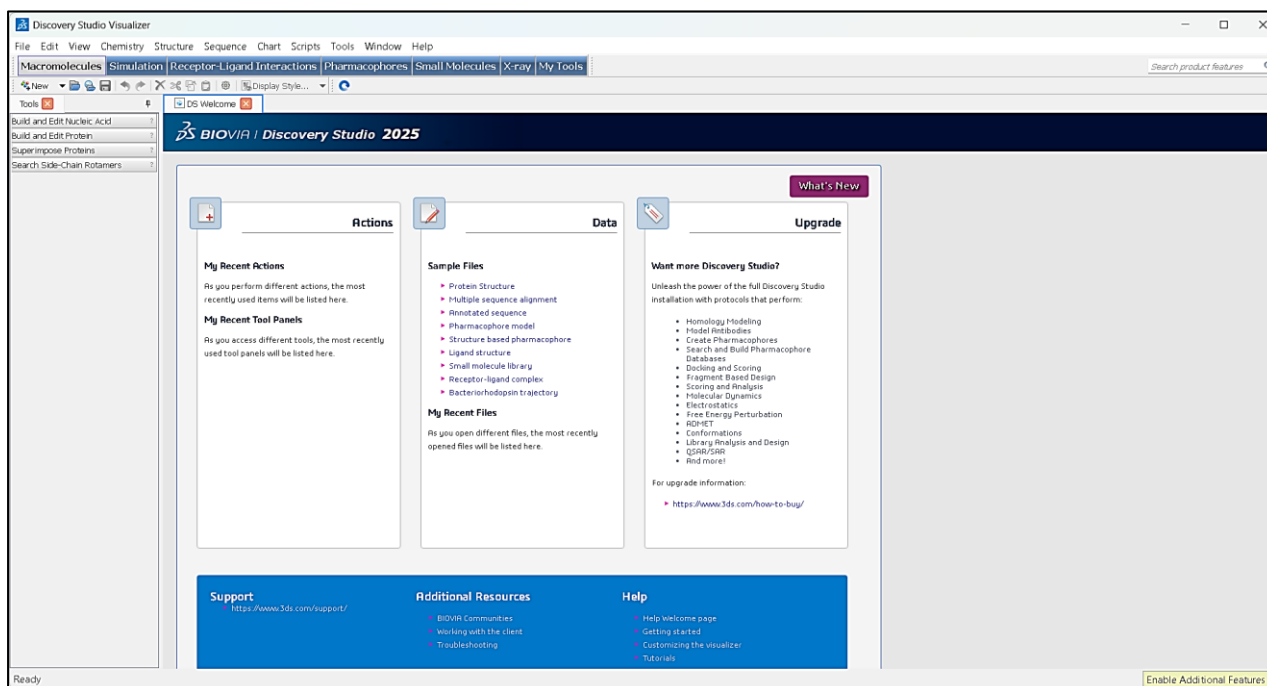


Fig.1: Homepage of Discovery Studio

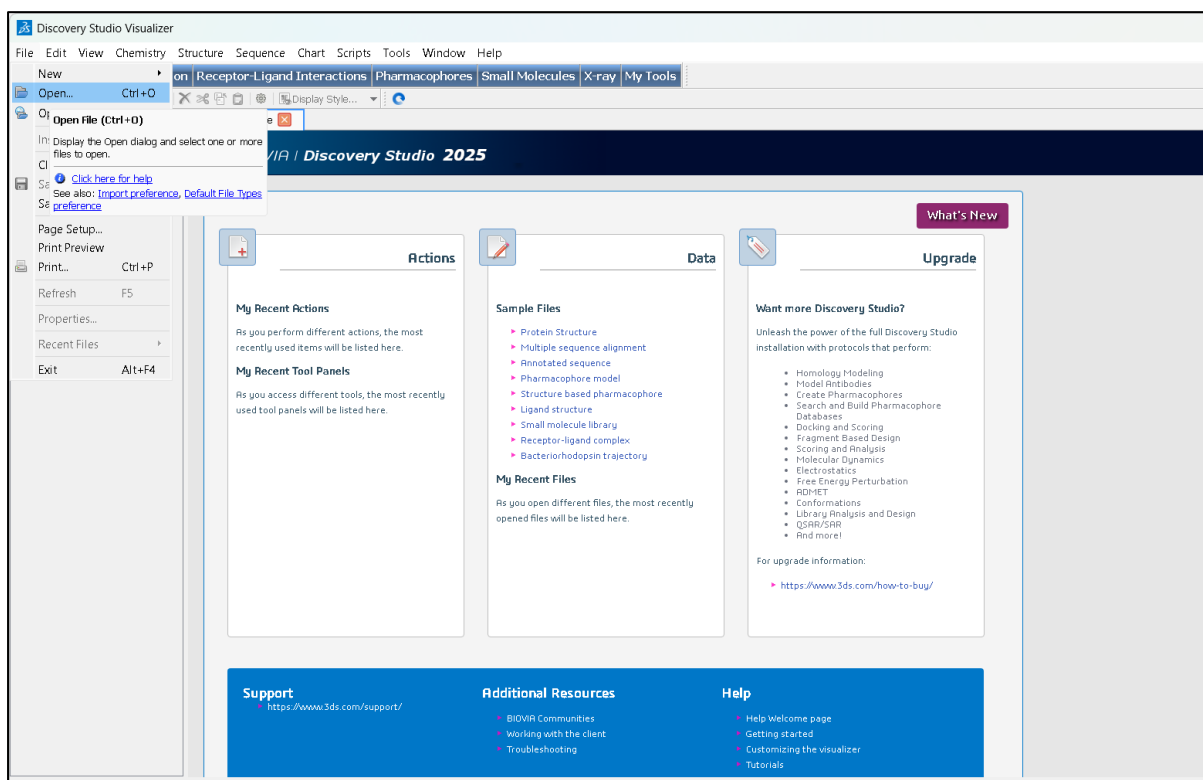


Fig.2: Go to file, and select open to load Lysozyme (1AKI) protein molecule

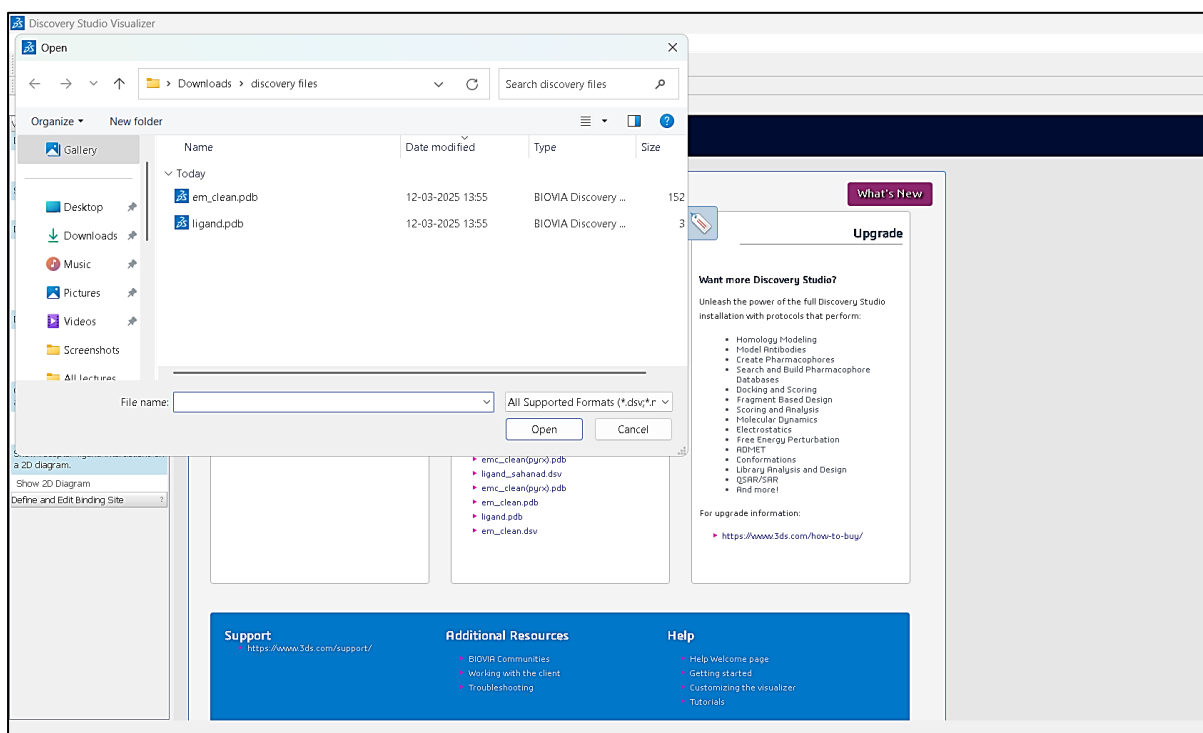


Fig.3: Protein molecule Lysozyme(Energy Minimized) and The lowest energy Ligand from PyRx

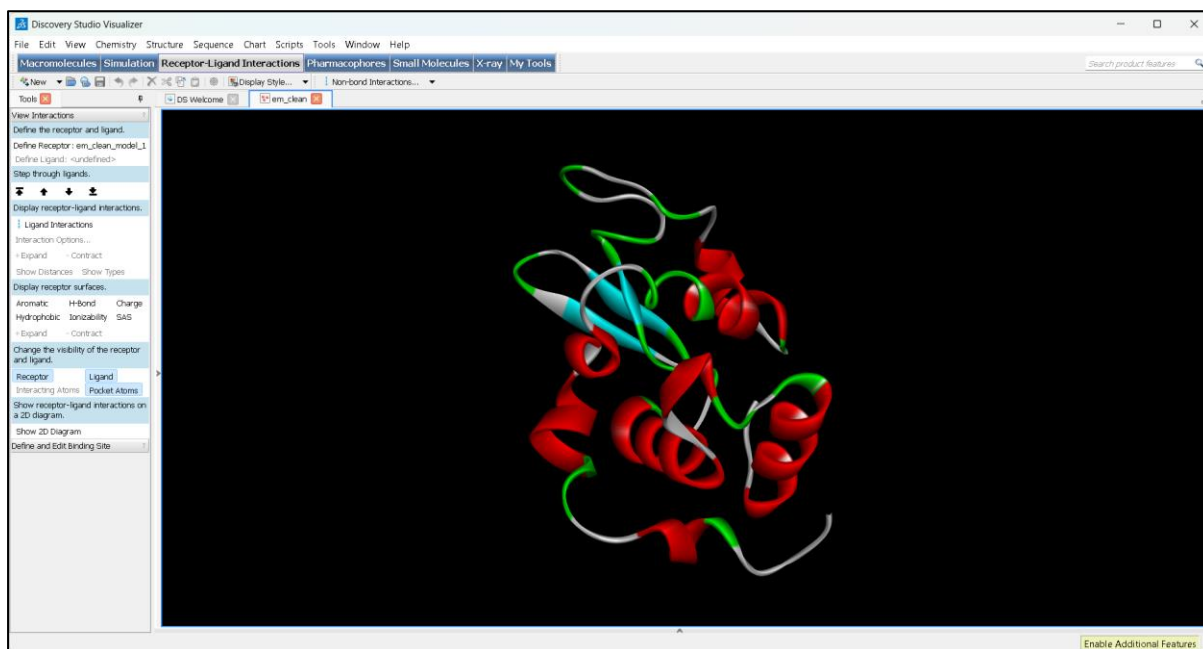


Fig.4: Loaded Lysozyme protein molecule

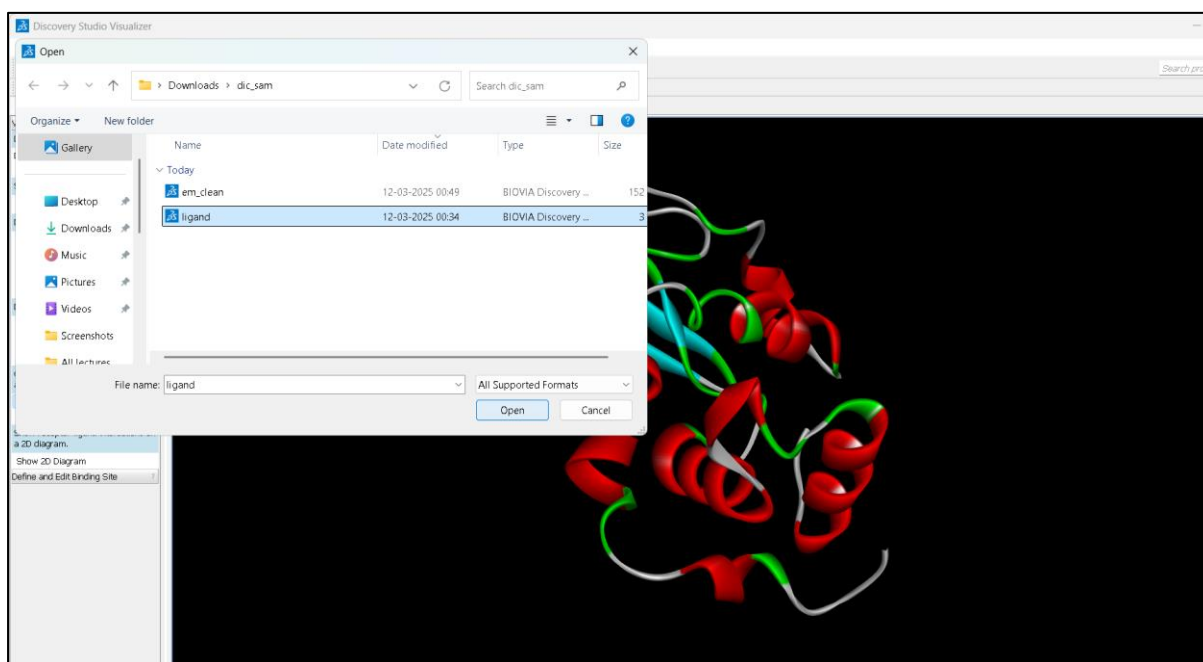


Fig.5: Loading Ligand

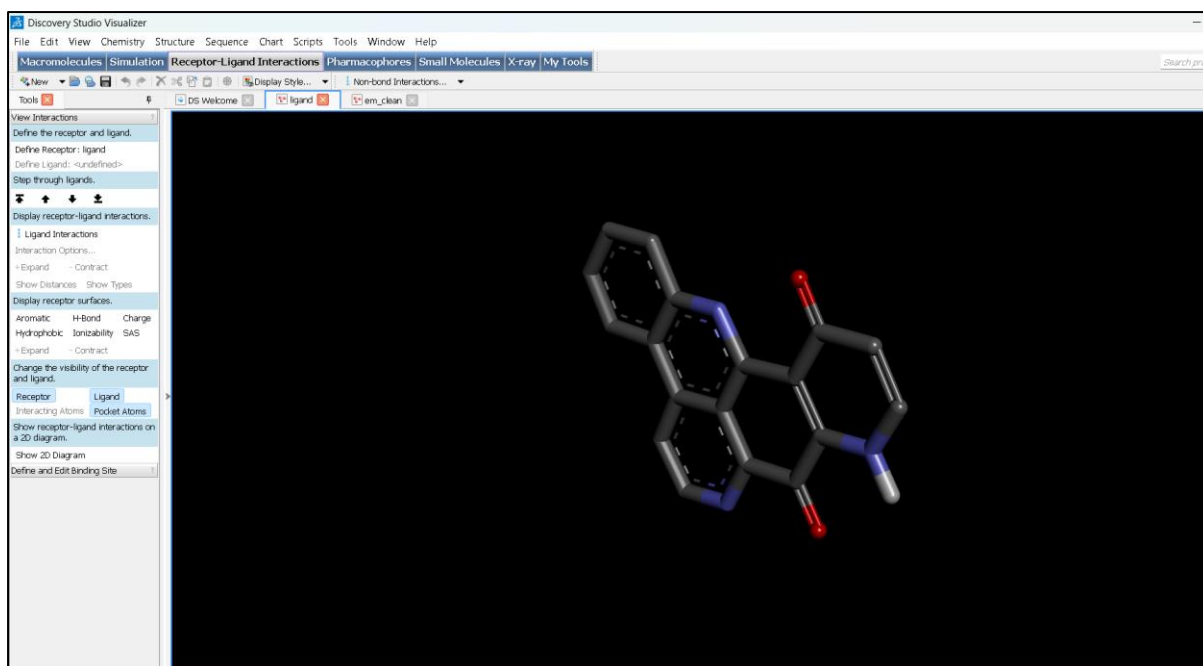


Fig.6: Ligand Loaded

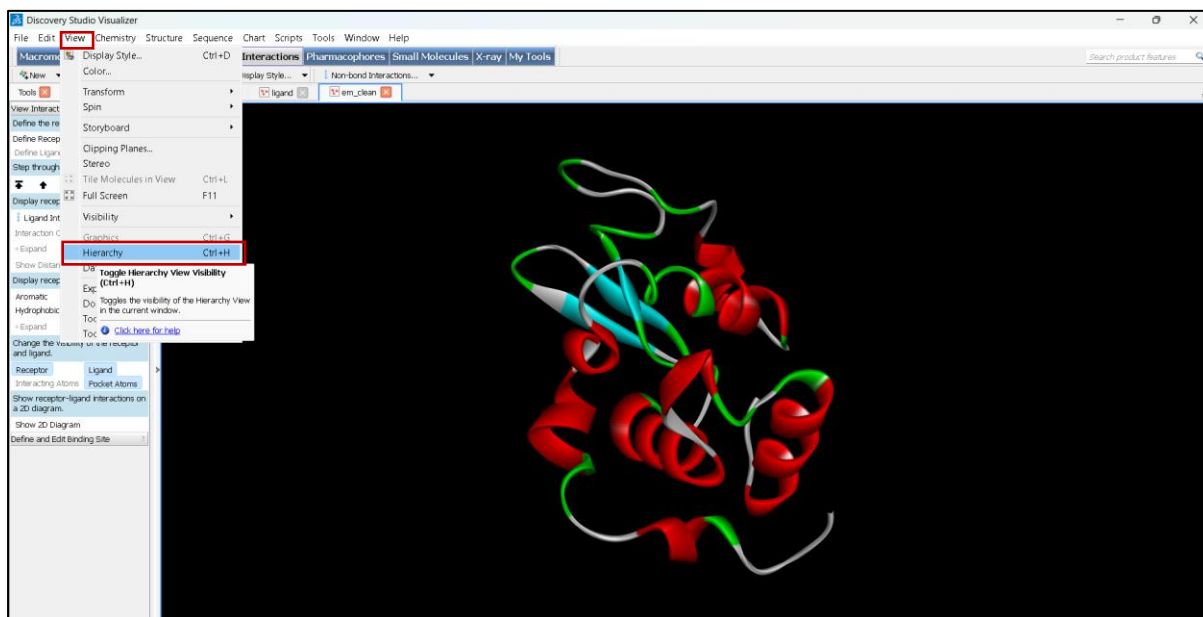


Fig.7: Selection of Hierarchy from view panel



Fig.8: Hierarchy section for the Lysozyme

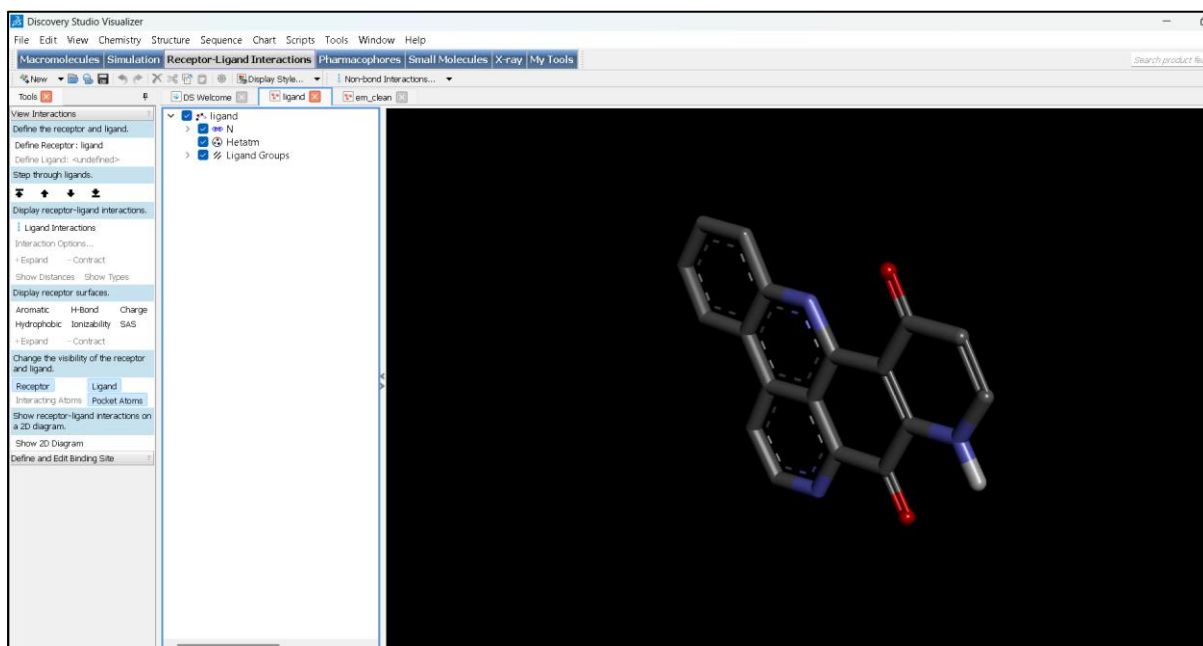


Fig.8: Hierarchy section for the Ligand (Meridine)

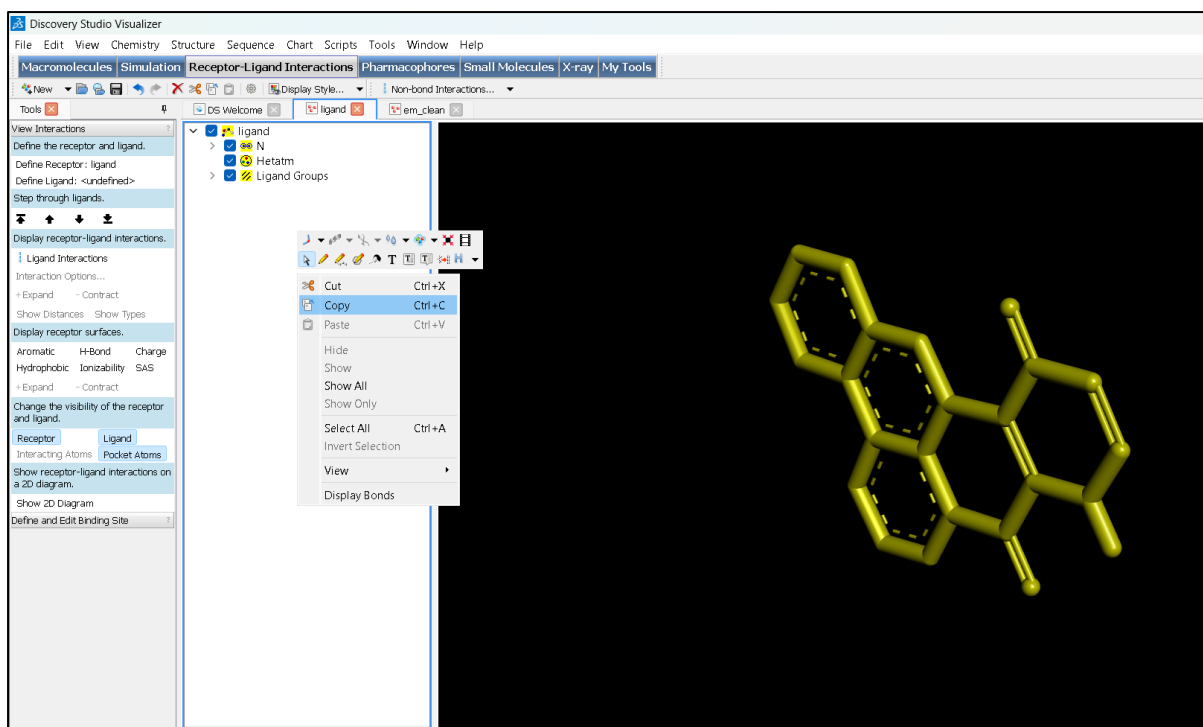


Fig.9: Copying the ligand molecule

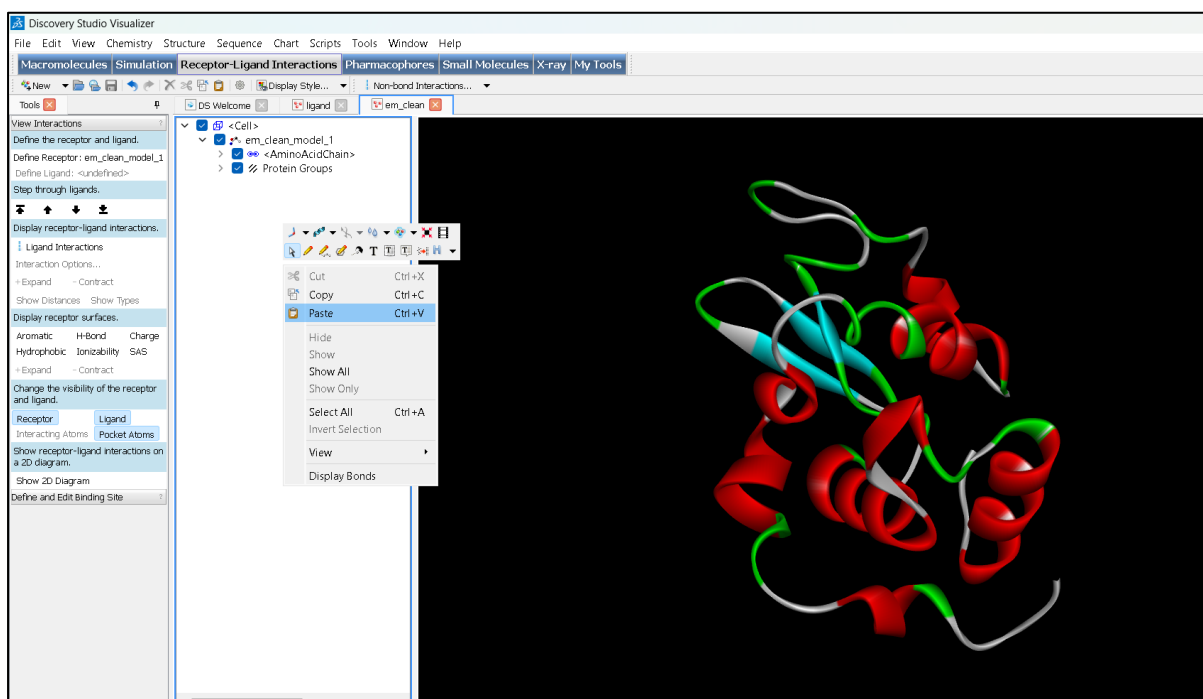


Fig.10: Pasting the ligand molecule in protein structure hierarchy

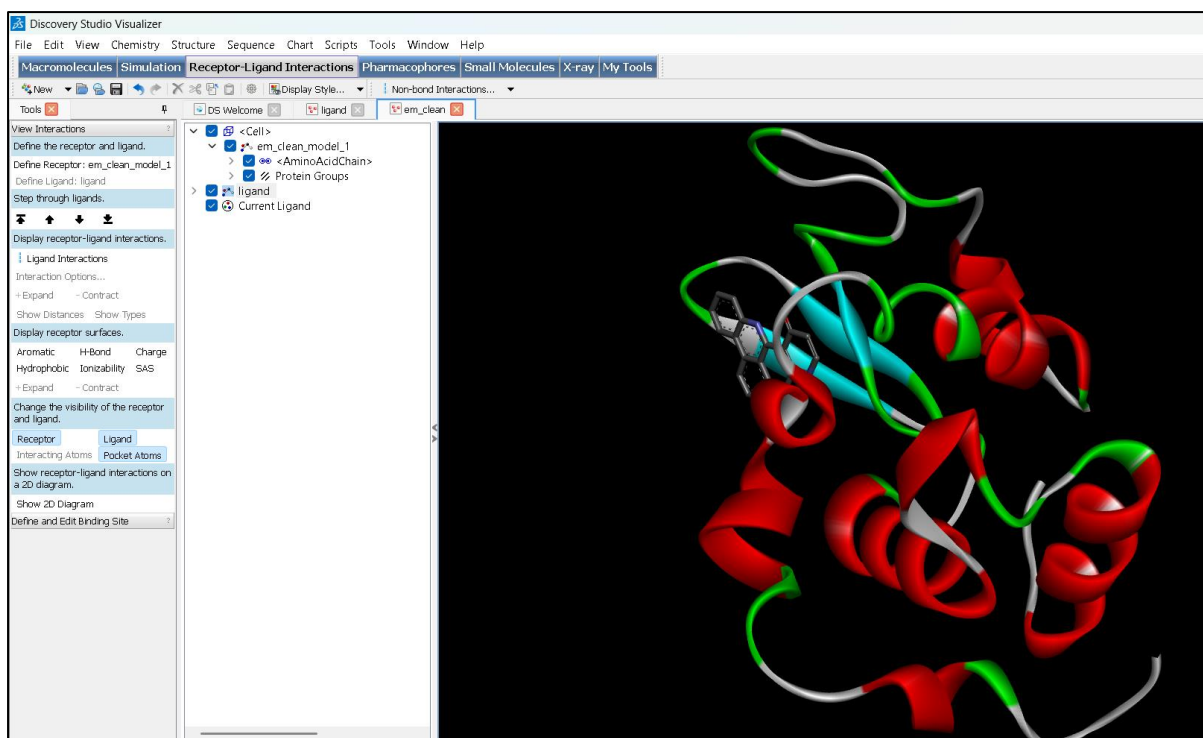


Fig.11: Ligand pasted

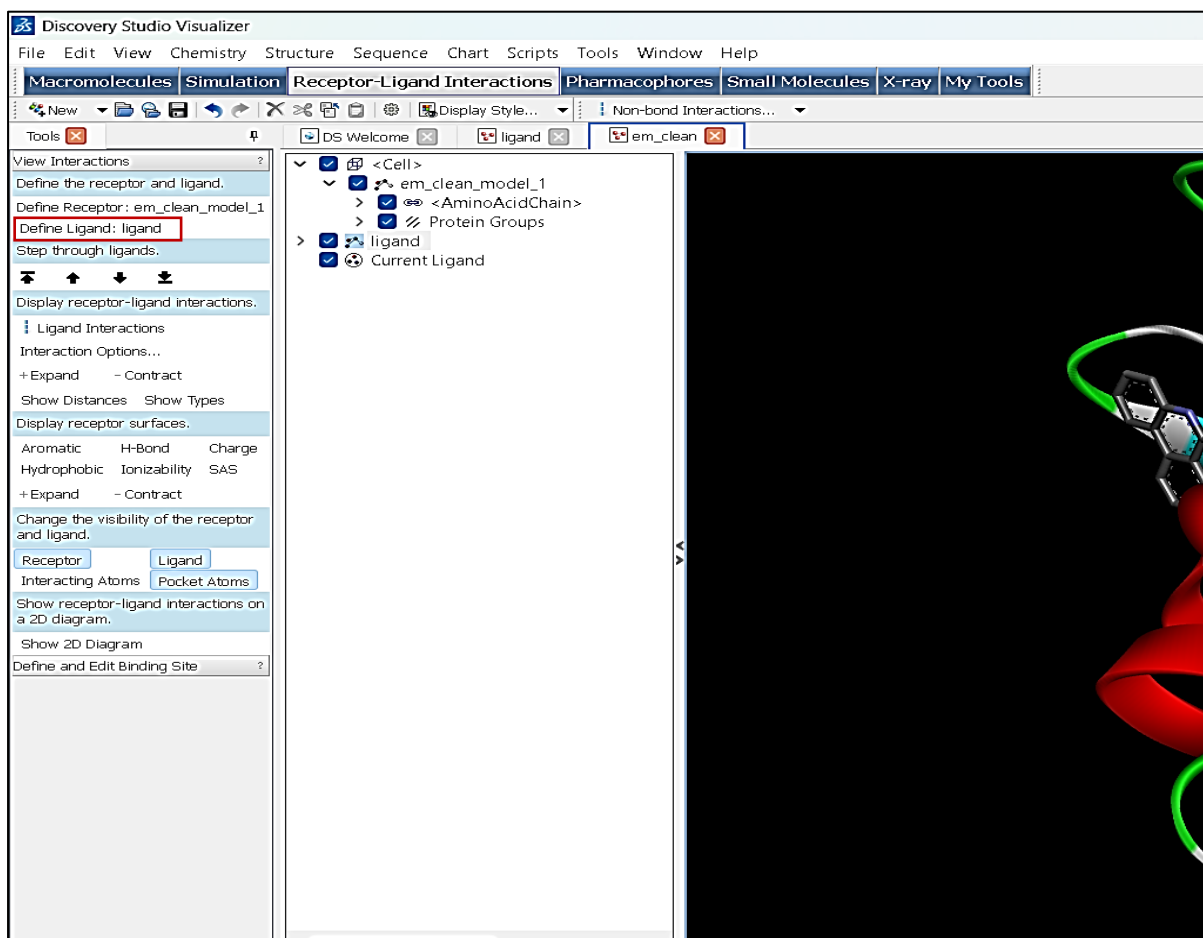


Fig.12: As the ligand is pasted, it gets defined

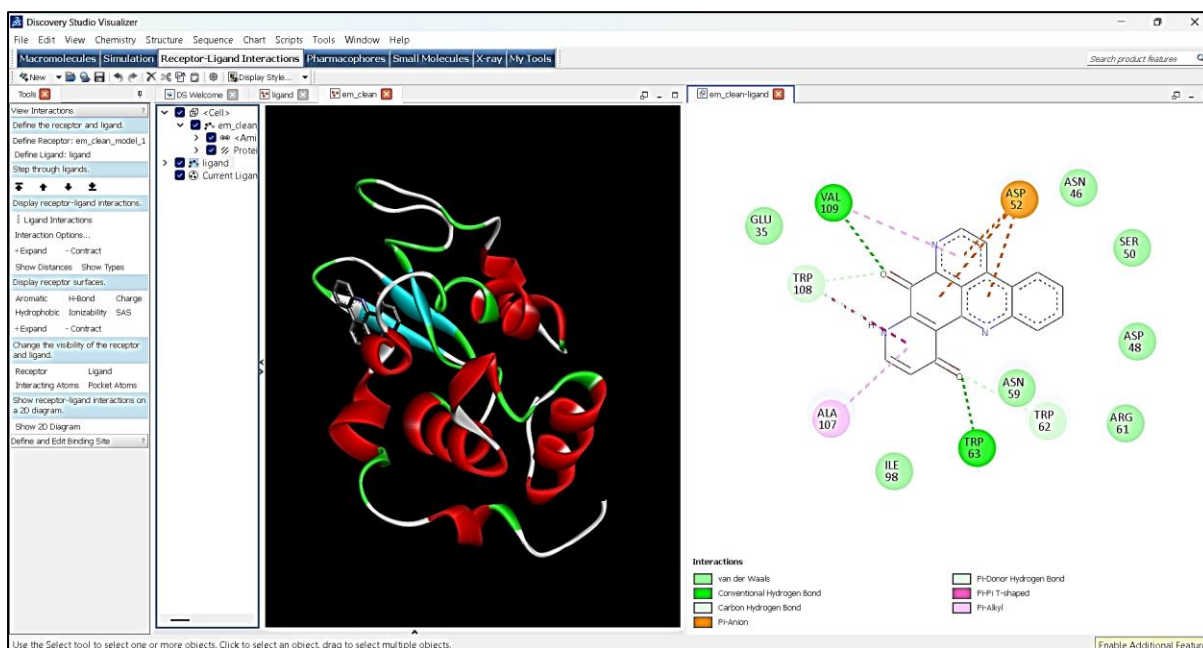


Fig.13: 2D and 3D Interaction of Lysozyme protein with phytochemical ligand Meridine

RESULTS:

Using the PyRx tool, the docked structure of Lysozyme (1AKI) protein and Meridine (CID: 72473) ligand was successfully obtained. A total of 9 conformers were generated, among which the best conformer (Model 1) was selected for further analysis and visualized in DS Visualizer. The 3D structure shows the receptor's α -helices (red), loops (green), and binding site residues (blue) interacting with the ligand, while the 2D interaction diagram highlights key molecular interactions: van der Waals forces (green), hydrogen bonds (light green), Pi-Anion interactions (orange), and Pi-Alkyl/Pi-Pi interactions (pink). Notably, ASP 52 forms Pi-Anion interactions, while TRP 63, TRP 108, ALA 107, and ILE 98 contribute hydrophobic interactions, stabilizing the ligand. Additional hydrogen bonds with ASN 59 and GLU 35 enhance binding.

CONCLUSIONS:

Discovery Studio Visualizer is a powerful tool for molecular modeling, docking, and visualization in drug discovery and bioinformatics. It enables researchers to analyze protein-ligand interactions, study binding sites, and predict molecular dynamics with high accuracy. The 2D interaction map was used to examine ligand-protein interactions in detail, highlighting key binding interactions. Additionally, the 3D structure of the docked complex was displayed, enabling rotation, zooming, and panning to explore the spatial orientation of the ligand within the binding pocket. The analysis revealed that Meridine fits snugly into the active site of Lysozyme, forming a stable complex, suggesting strong binding affinity. By integrating computational chemistry, docking studies, and structural biology, Discovery Studio enhances drug design, biomolecular research, and structure-based predictions.

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<https://pubchem.ncbi.nlm.nih.gov/compound/Meridine>
 2. BIOVIA. (n.d.). BIOVIA DISCOVERY STUDIO®. In Datasheet.
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